Image Processing Part 2

CM 515 - Rosi Danzman and Kelsey Martin

Images in Research

Images contain copious information

Images can be used to answer many questions

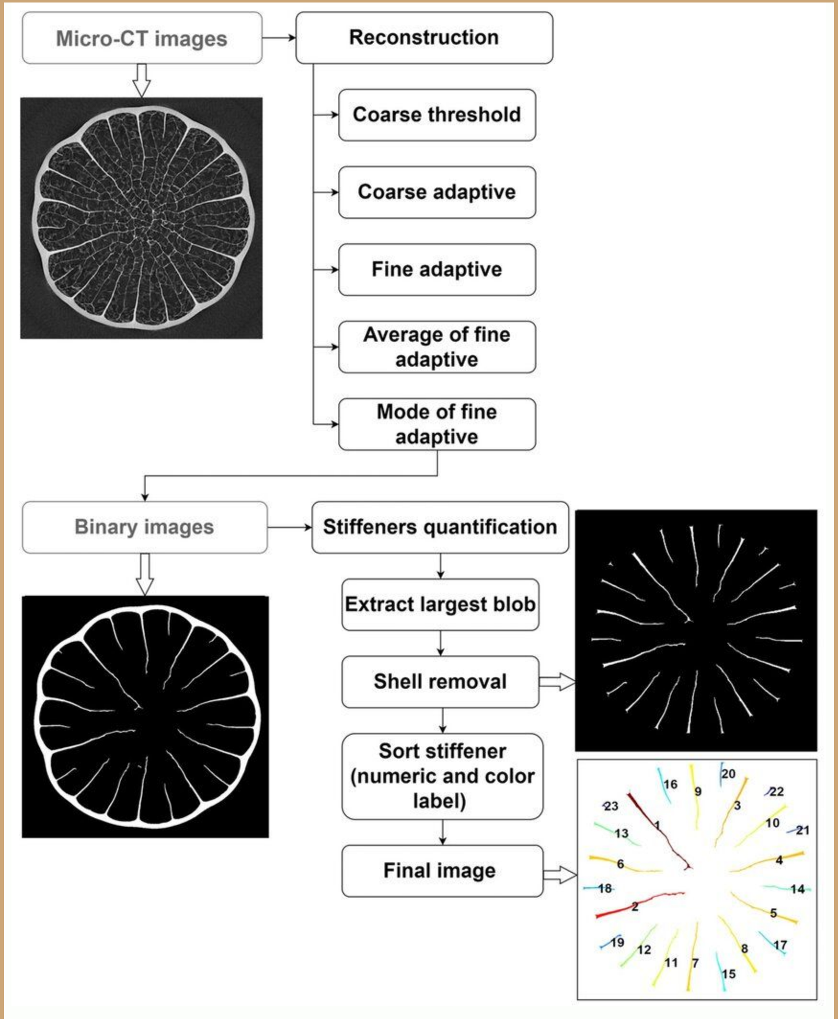
What is the area of this nucleus?

How many seeds are present?

What stage of the cell cycle are these cells at?

Objectives	General Examples (Descriptive Semantics)	Analytical Metrics (Quantitative Semantics)	Common Tools
Characterizing biological motion	Organelle movement, cell locomotion	Velocity, directionality, persistence	Object tracking
	Saltatory movement or processivity of motor-driven movement	Velocity, directionality, persistence, diffusion constant	Particle tracking
	Molecular turnover in a compartment	Dissociation constant, diffusion coefficient	FRAP, photoconversion
Interpreting biological interaction/signaling	Target protein in an organelle	Co-occurrence coefficient, correlation coefficient	Colocalization analyses
	In situ levels of protein modifications (e.g. phosphorylation)	Image ratio	Ratiometric imaging
	Microenvironment of target molecules	Anisotropy	Anisotropy measurement
Measuring shape/size of biological structures	Aspect ratio during cell spreading or cell differentiation	Shape descriptors	Morphometry
	Filopodial dynamics, membrane ruffles in 3D	Turnover, angular deflection, surface curvature	Deformation measurements

Example of image processing?

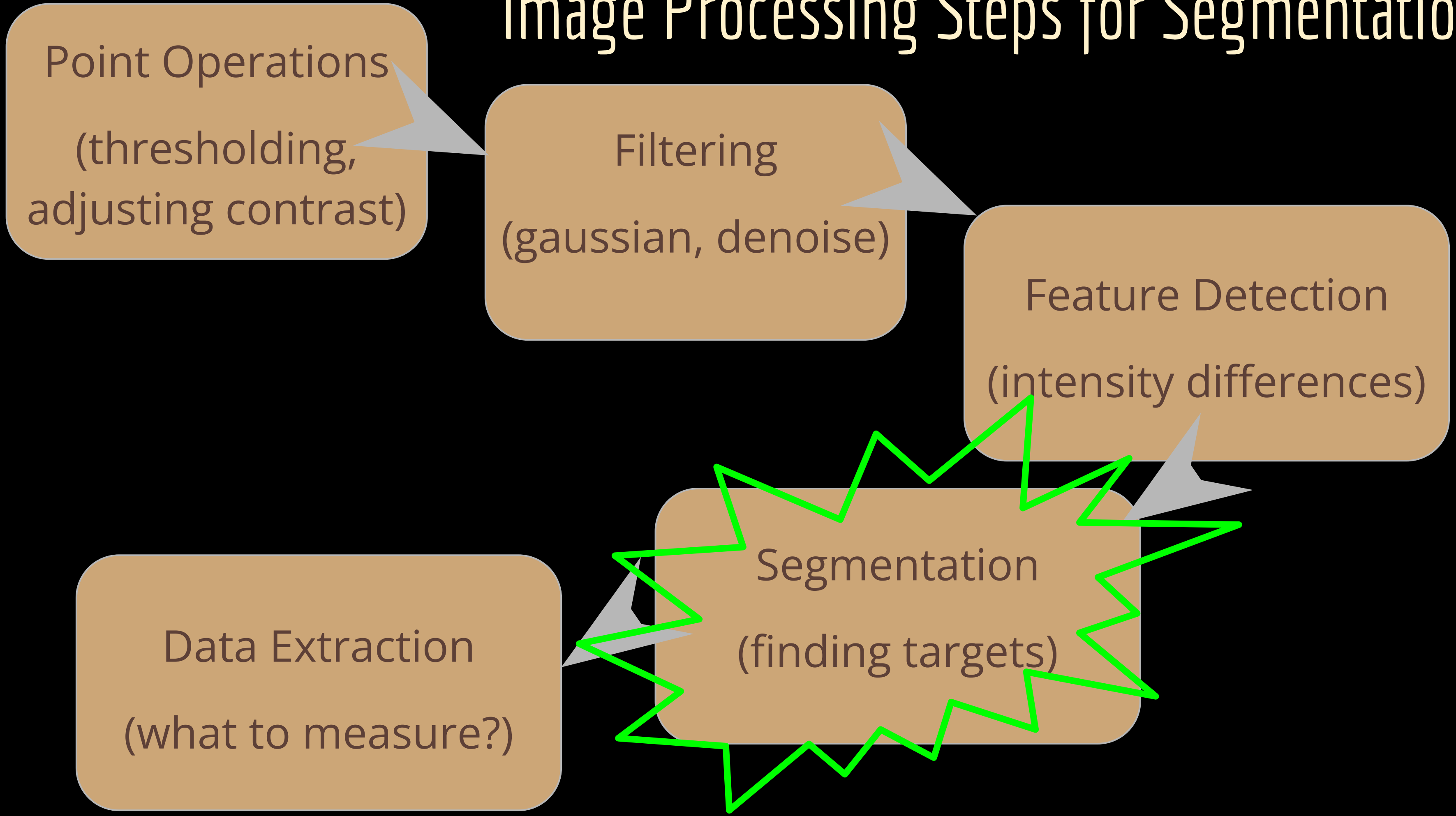



Huawei P30 Pro (Auto)
Samsung Galaxy S10+ (Auto)
Google Pixel 3 (Auto)
Huawei P30 Pro (Night)
Samsung Galaxy S10+ (Night)
Google Pixel 3 (Night)

NEVER MESS WITH RAW IMAGES

Always make a copy of your raw images to work with

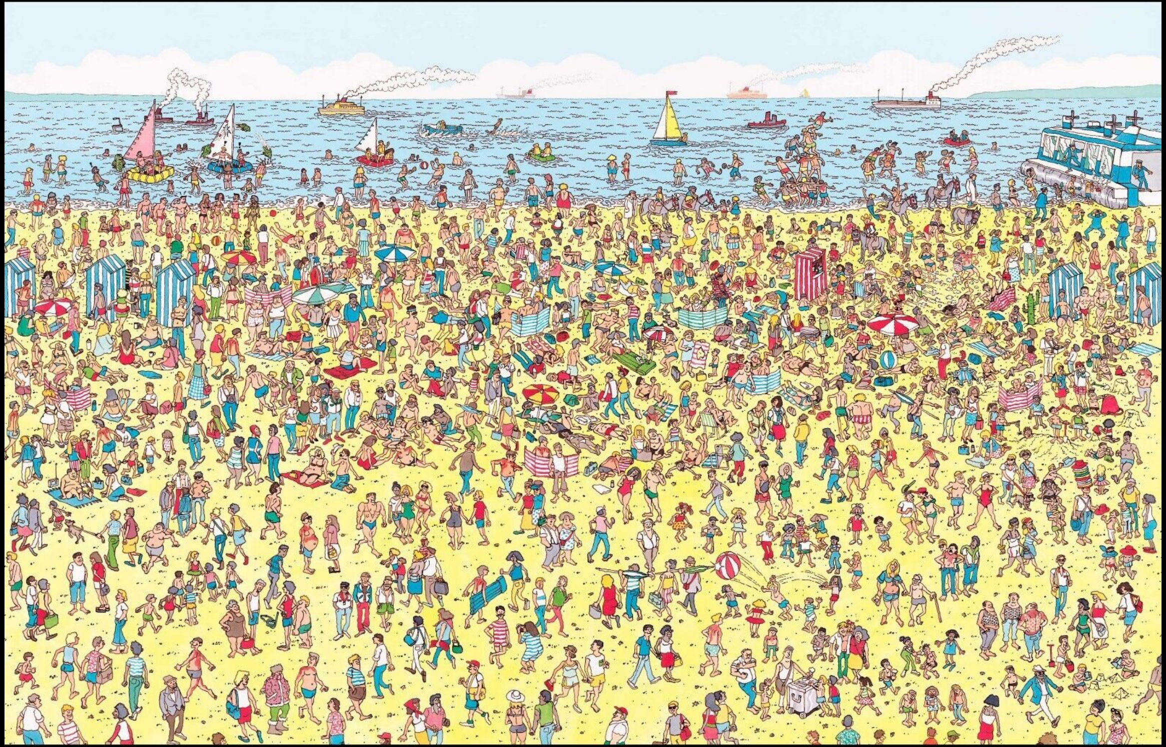
Image Processing Steps for Segmentation



Segmentation

Segmentation - finding a target

Must identify boundaries to measure something



Computers use similarities and discontinuities

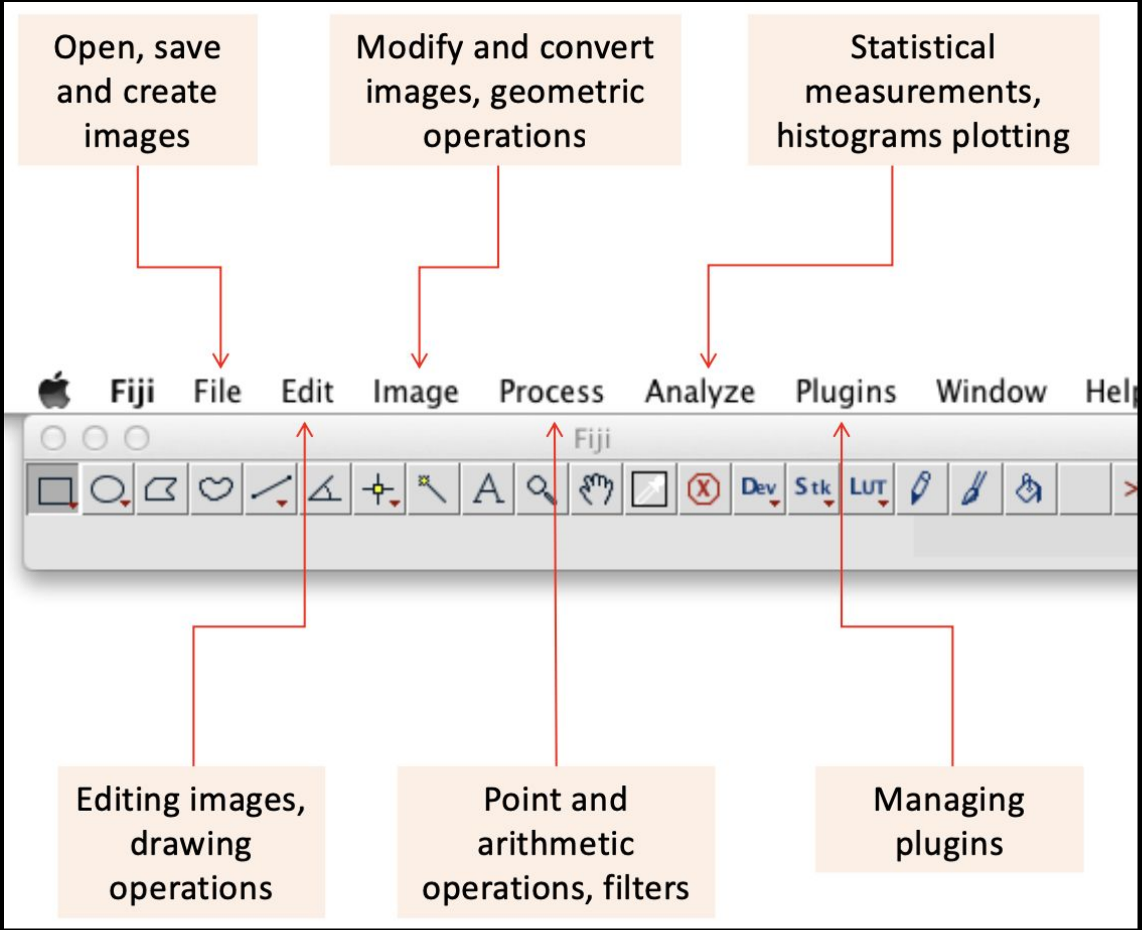
Ex: thresholding or edge detection

How segmentation is done

Humans can easily pick out parts of images - segment - computers need help

Open FIJI

→ File
→ Open Samples
→ hela-cells…



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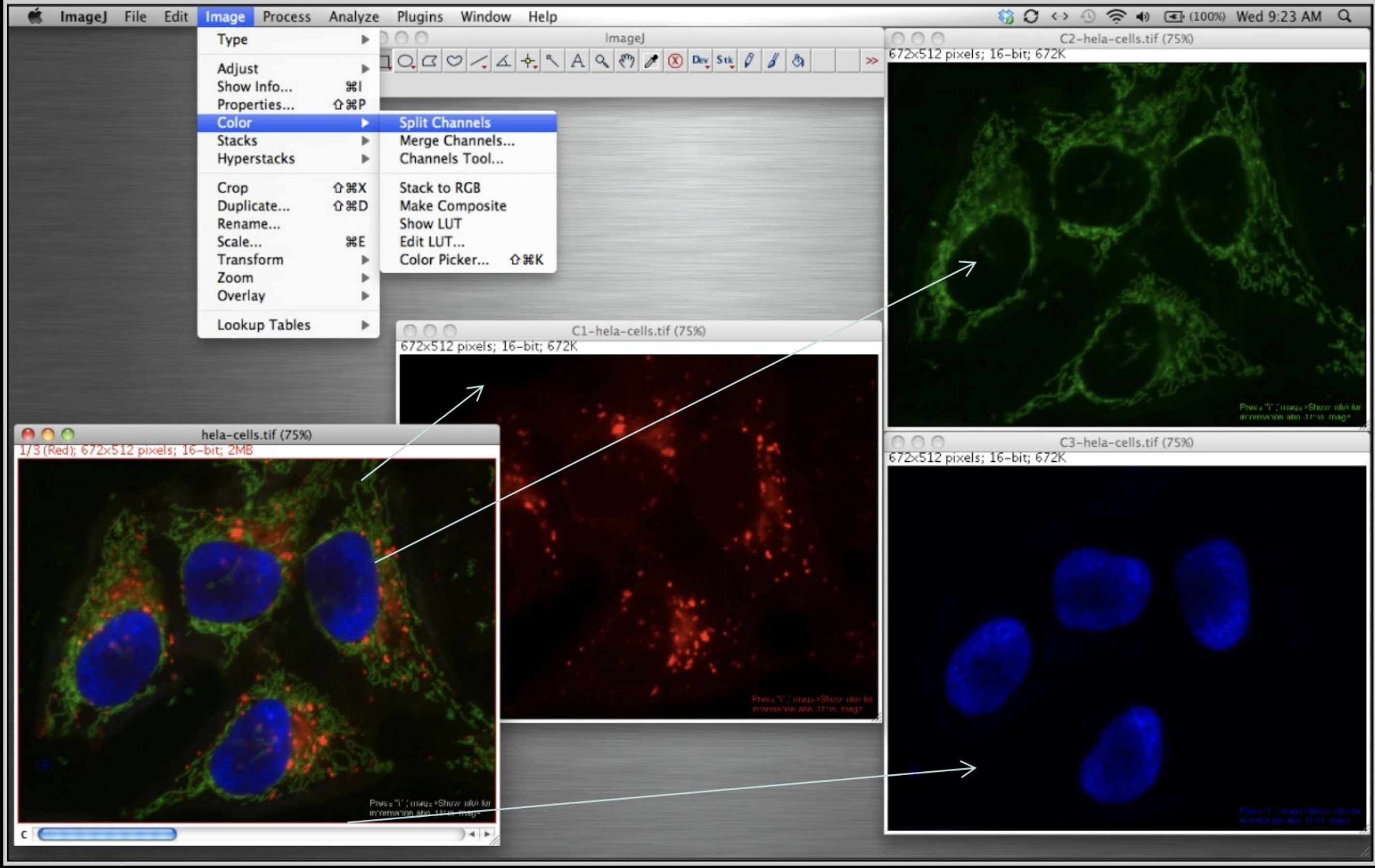
Let's Split Channels

Convert to 8-bit
→Image
→Type
→ 8bit

Split channels
→Image
→Color
→Split Channels

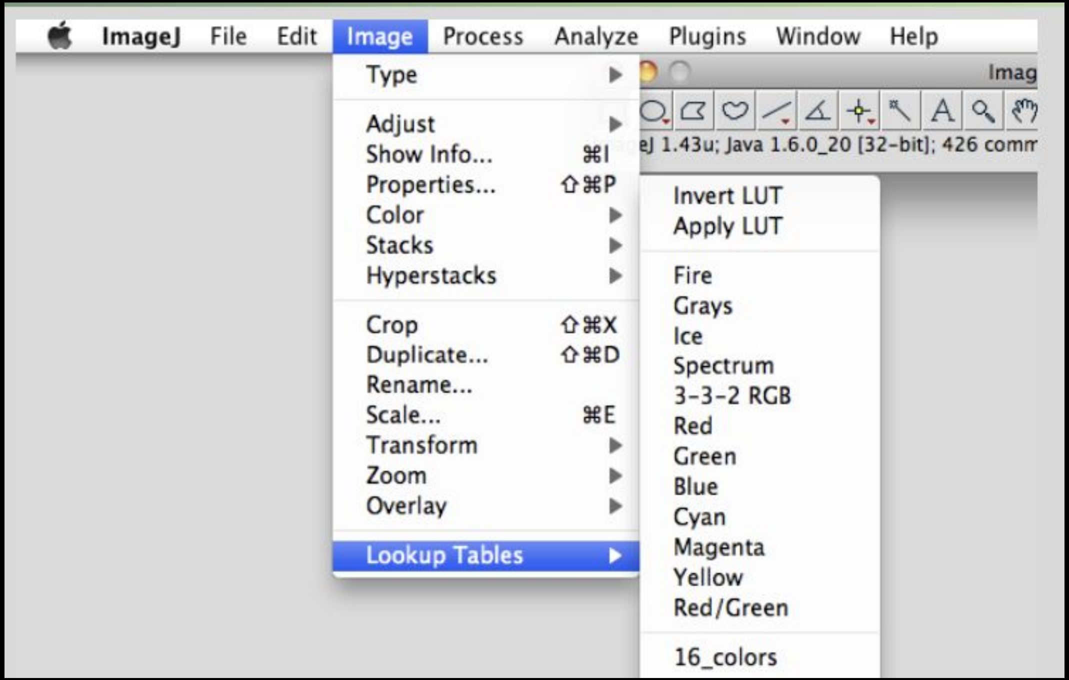
Save each as .png



Look Up Tables (LUT)

Try a few on each image

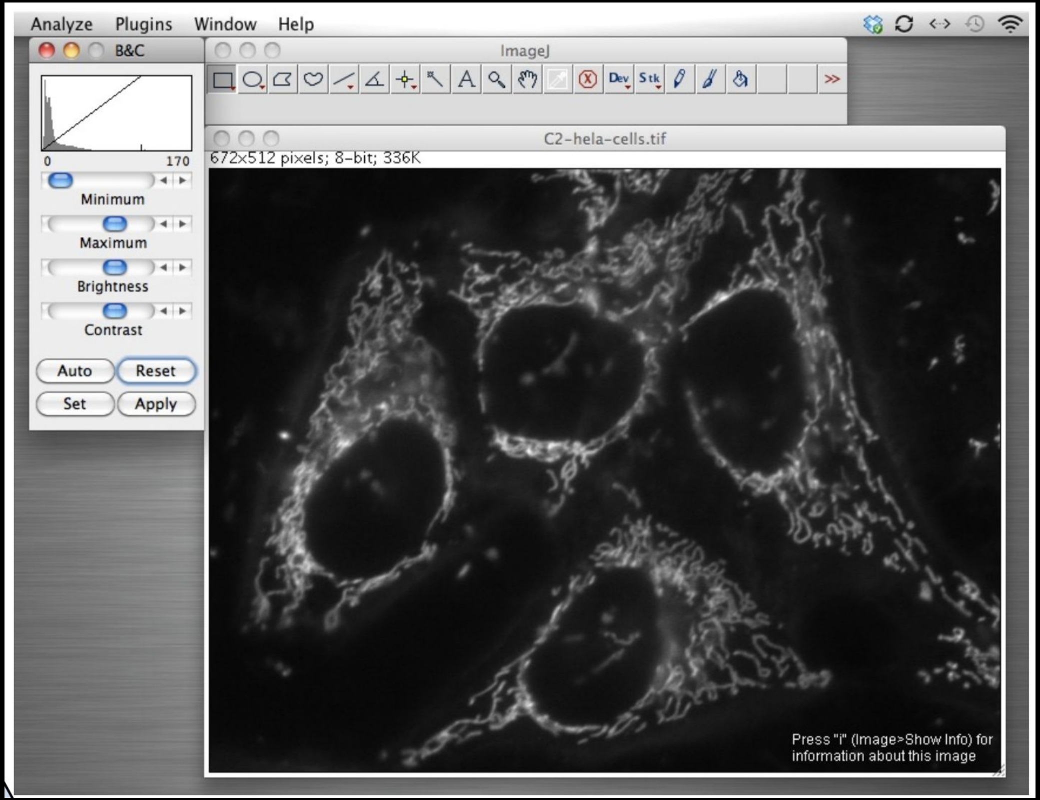
LUT do not alter pixel values, just display



Brightness and Contrast

Shift + cmd + c

This 'shrinks' the data displaying power - changes pixel value of display but not ratios



Merge channels ⟶Image

Point Operations

Point Functions change each pixel based only on that pixels original value…

These functions have limited functionality - we need more tools to reach our goal … segmentation

Filtering

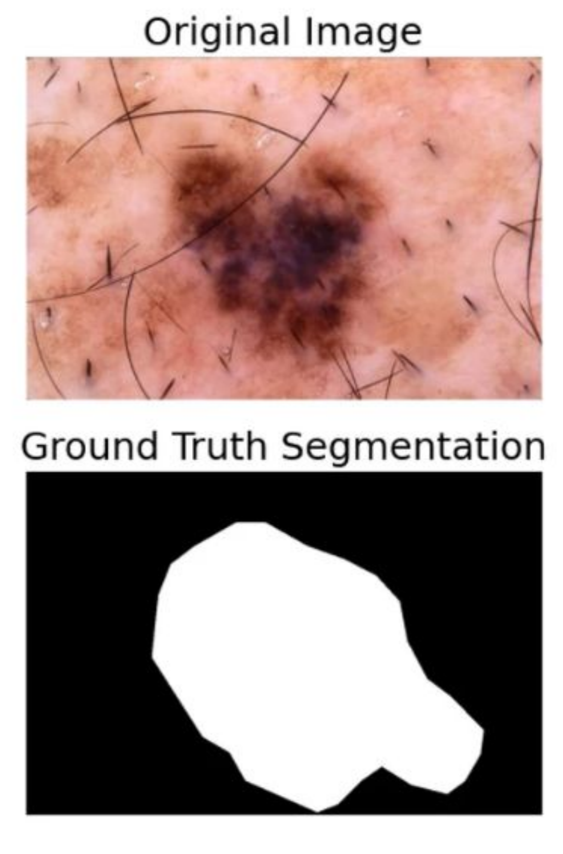
Filters are algorithms performed on kernels (group of pixels)

Enhancing features for segmentation - before data extraction

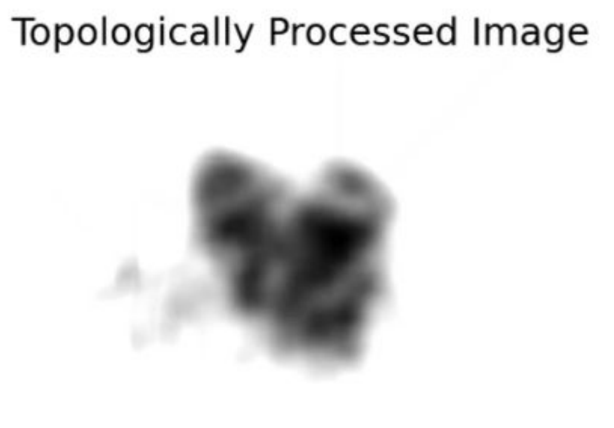
This alters your data (pixel values)! Cannot use altered images for quantification - they are only for setting up segmentation

Filtering enhances the signal

Skin lesion photograph

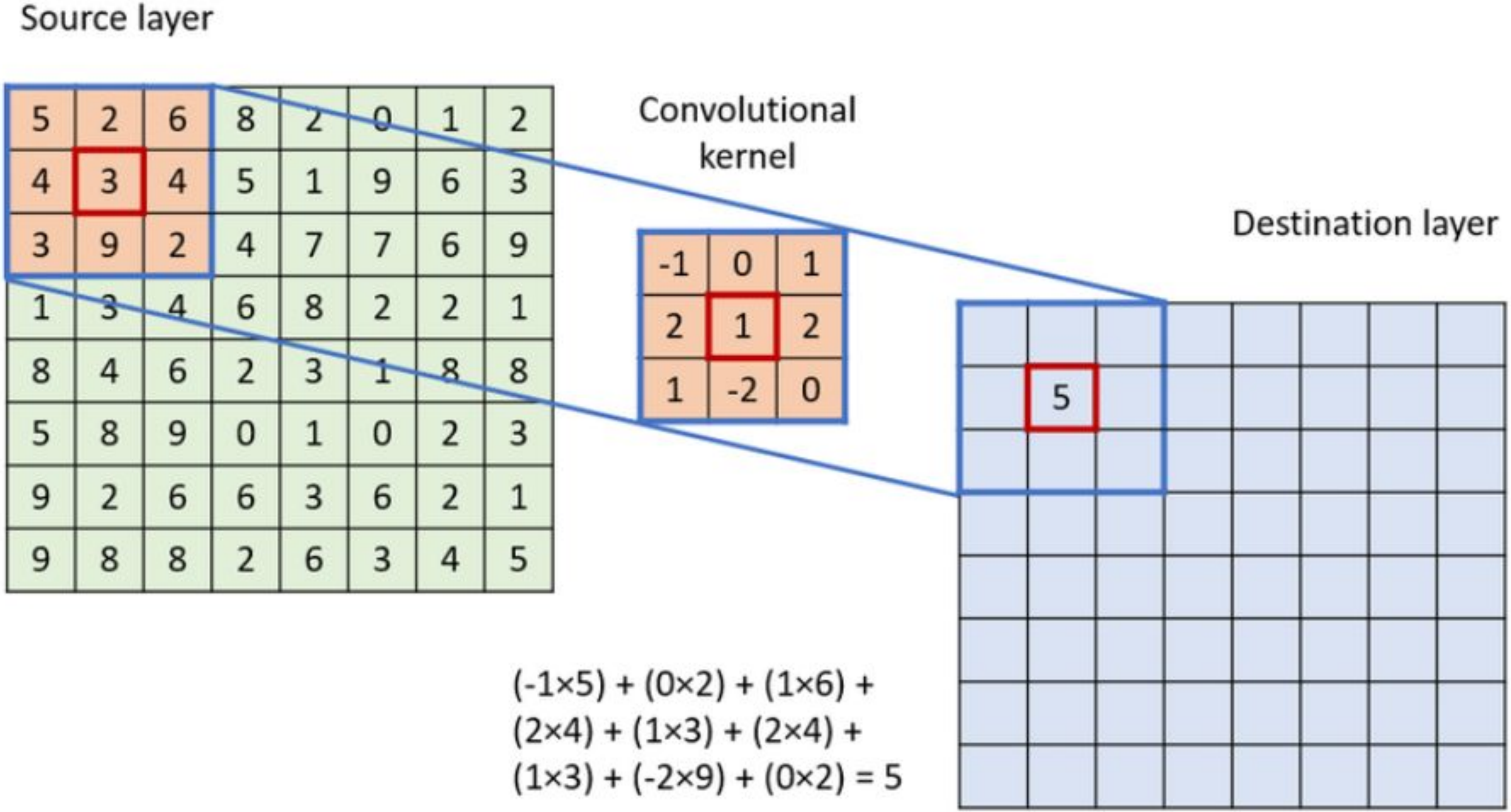


Vandaele et al., 2020



MODIFIED IMAGES CANNOT BE USED FOR ANALYSIS

Messing with images changes the data - they can only be used for display or processing...to prep for analysis


Source layer
Convolutional kernel
Destination layer
5
(-1×5) + (0×2) + (1×6) +
(2×4) + (1×3) + (2×4) +
(1×3) + (-2×9) + (0×2) = 5

Filter Kernels

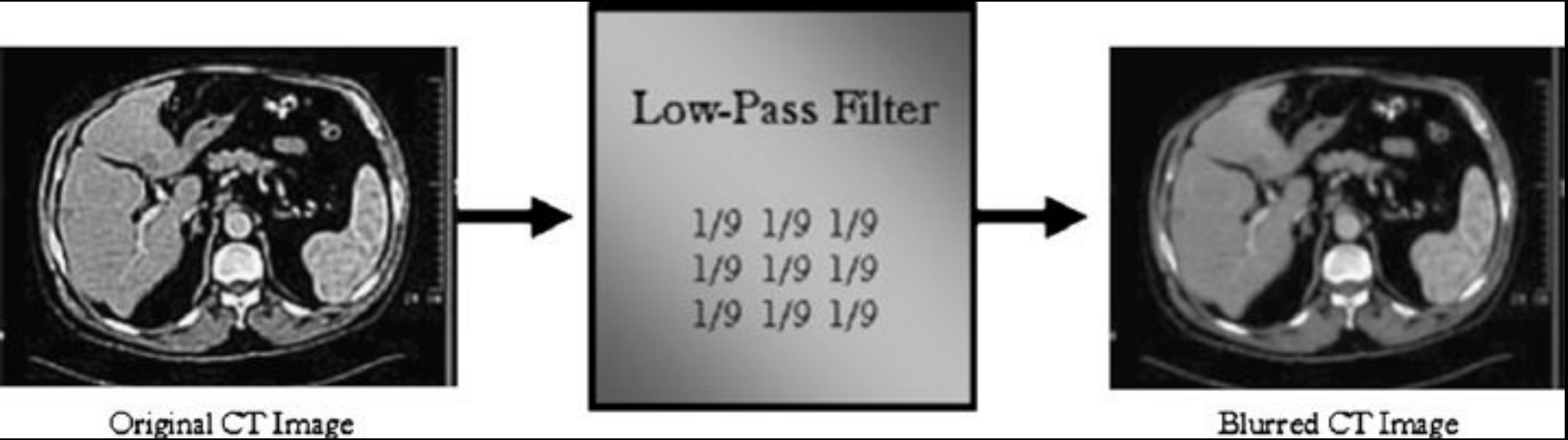
Original	*Gaussian Blur*	*Sharpen*	*Edge Detection*
$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$
			

Smoothing Filters (low-pass)

Leaves (passes) low spatial frequencies untouched

High frequencies are attenuated, almost absent

Frequency - refers to rate of change in pixel values



Low pass filter - smoothing

High Frequency removed (fine features or rapid changes)

Smoothing in FIJI

Download coins.tif from GitHub module

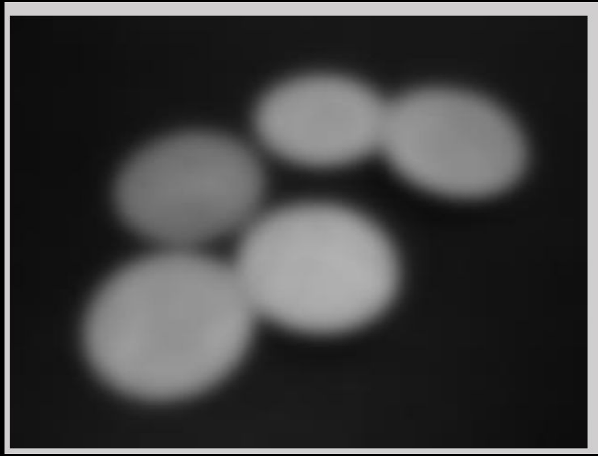
Test smoothing filters:

→ Process

→ Smooth

→ Filters → Gaussian

→ Median



Gaussian Blur Filter

Kernel is bell shaped (gaussian curve)

Kernel = 2 * sigma + 1
(you set the sigma)

Center is heavily weighted

Sigma = 7 (15X15)

2	2	3	4	5	5	6	6	6	5	5	4	3	2	2
2	3	4	5	7	7	8	8	8	7	7	5	4	3	2
3	4	6	7	9	10	10	11	10	10	9	7	6	4	3
4	5	7	9	10	12	13	13	13	12	10	9	7	5	4
5	7	9	11	13	14	15	16	15	14	13	11	9	7	5
5	7	10	12	14	16	17	18	17	16	14	12	10	7	5
6	8	10	13	15	17	19	19	19	17	15	13	10	8	6
6	8	11	13	16	18	19	20	19	18	16	13	11	8	6
6	8	10	13	15	17	19	19	19	17	15	13	10	8	6
5	7	10	12	14	16	17	18	17	16	14	12	10	7	5
5	7	9	11	13	14	15	16	15	14	13	11	9	7	5
4	5	7	9	10	12	13	13	13	12	10	9	7	5	4
3	4	6	7	9	10	10	11	10	10	9	7	6	4	3
2	3	4	5	7	7	8	8	8	7	7	5	4	3	2
2	2	3	4	5	5	6	6	6	5	5	4	3	2	2

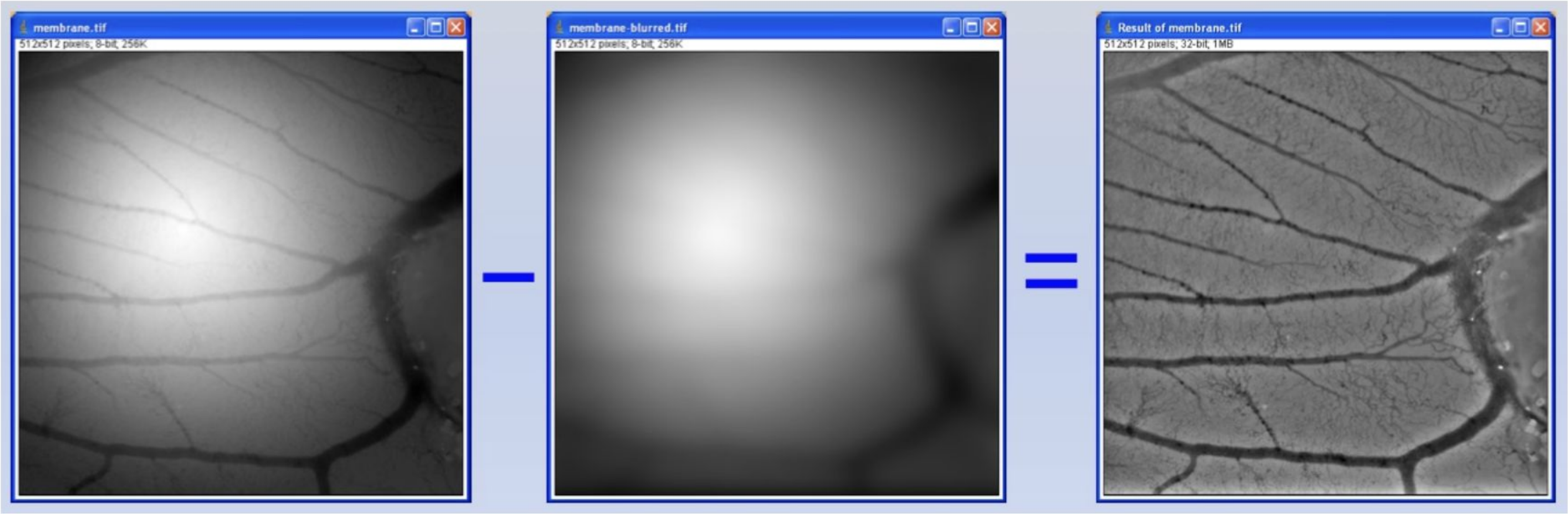
Sigma = 3 (7X7)

1	1	1	2	1	1	1
1	2	2	4	2	2	1
2	2	4	8	4	2	2
2	4	8	16	8	4	2
2	2	4	8	4	2	2
1	2	2	4	2	2	1
1	1	1	2	1	1	1

Sigma = 2 (5X5)

1	1	2	1	1
1	2	4	2	1
2	4	8	4	2
1	2	4	2	1
1	1	2	1	1

Why would you BLUR!!

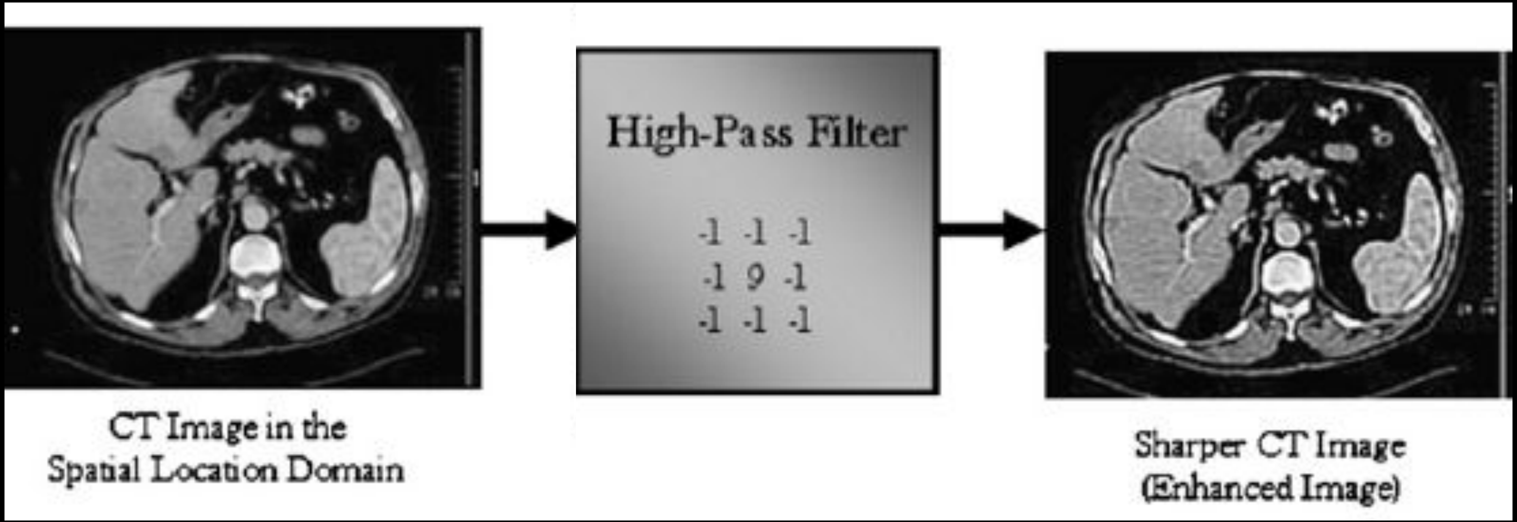


Sharpening (High-Pass Filter)

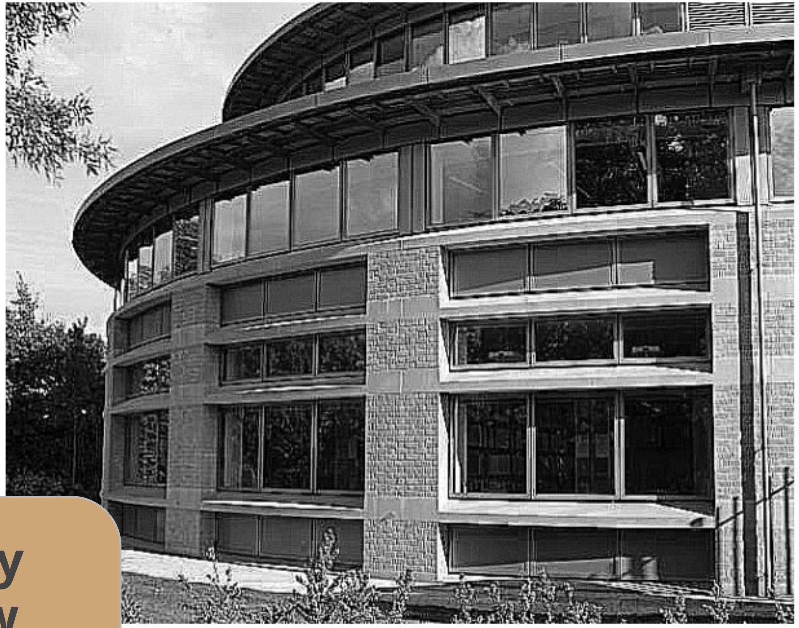
High Frequency = rapid changes, fine texture, details

Passes up the high spatial frequency pixels,

Attenuates low frequencies…this accentuates high frequencies



High-pass filter - sharpening



Low Frequency removed (slow changes or smooth bits)

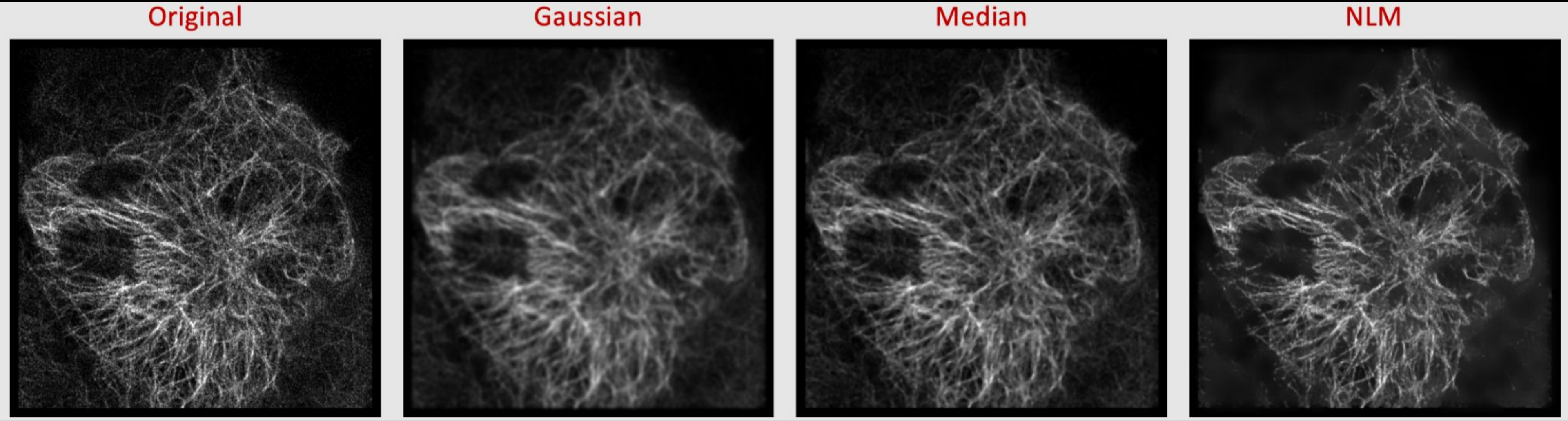
Denoising - smoothing while keeping edges

Neighborhood methods

- Uses surrounding pixel values
- Mean, median, gaussian
- Performs poorly on noisy images

Non-Local methods

- Whole image is scanned by kernels
- Averages similar pixels to reduce noise
- Considers similarities of pixel values across whole image



Non-Linear Filters

Uses a nonlinear function to modify pixels → uneven modifications (be careful!)

Segmentation

Picking out objects of interest

Point functions and filters help the computer see the objects

But we still need tools to find those objects and remove the other stuff

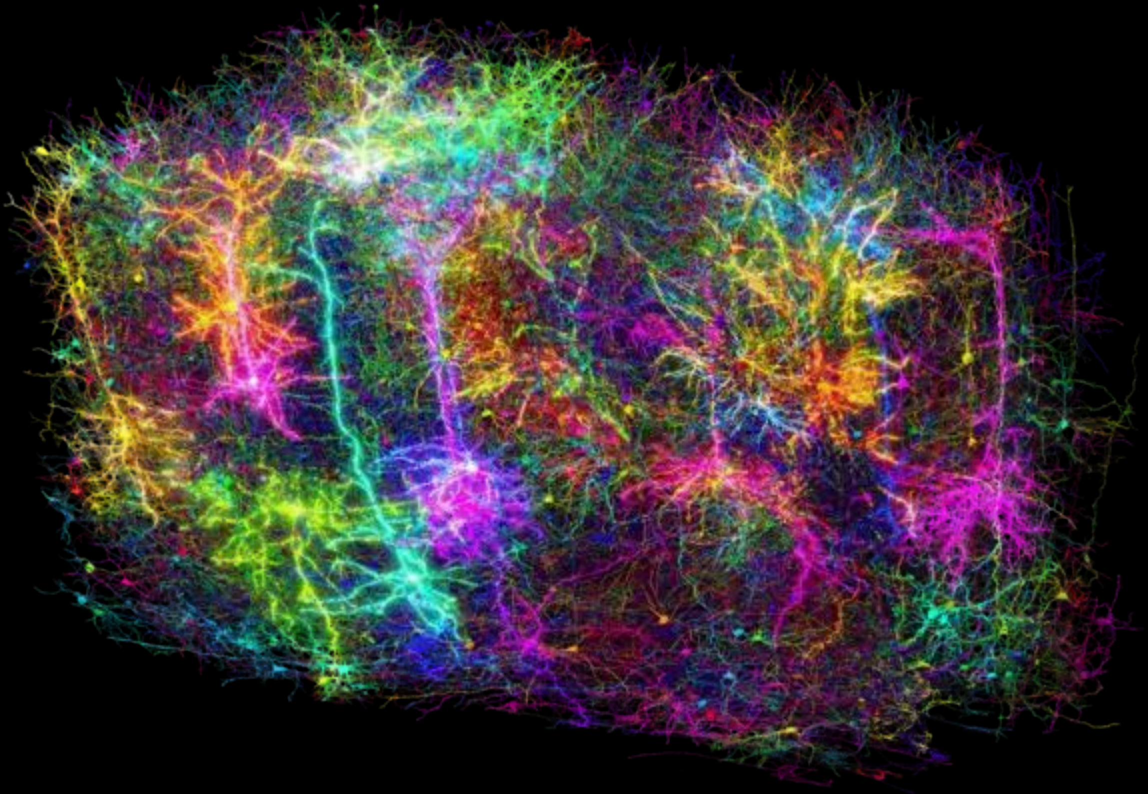


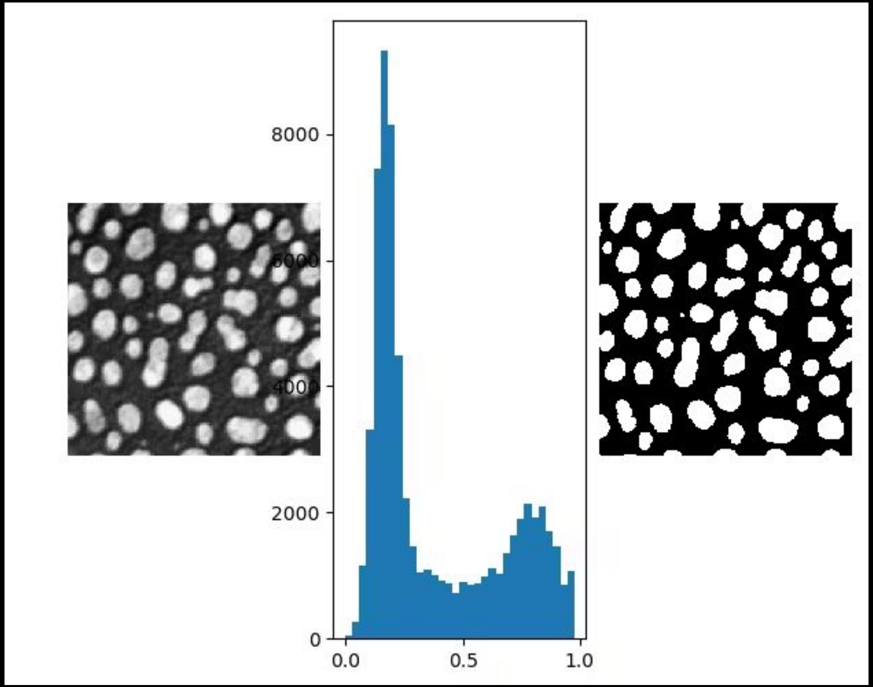
Image credit: Forrest Collman, Allen Institute

Thresholding Method for segmentation

Setting a cut off to exclude non-object pixels

similar pixel groups ⟶ dark or bright

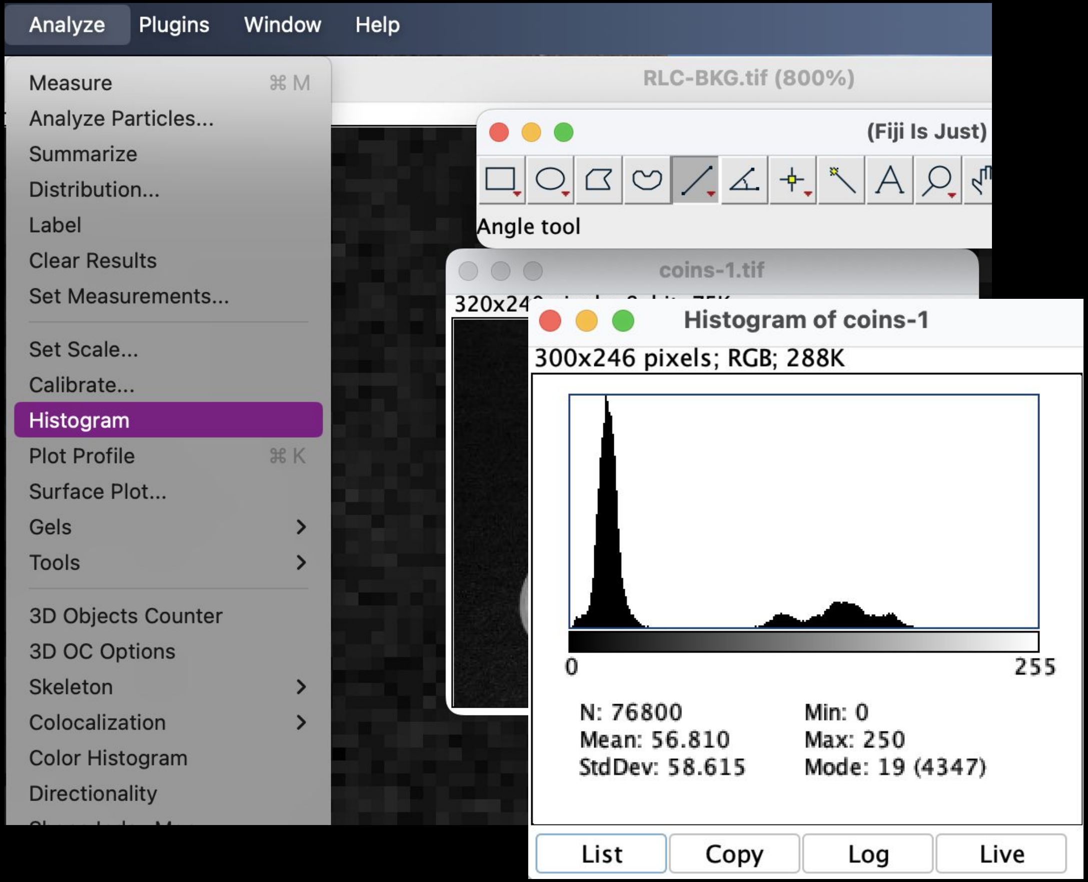
There are other methods we will not cover



Looking at the pixel values

Open coins.tif from GitHub

→ Analyze → Histogram

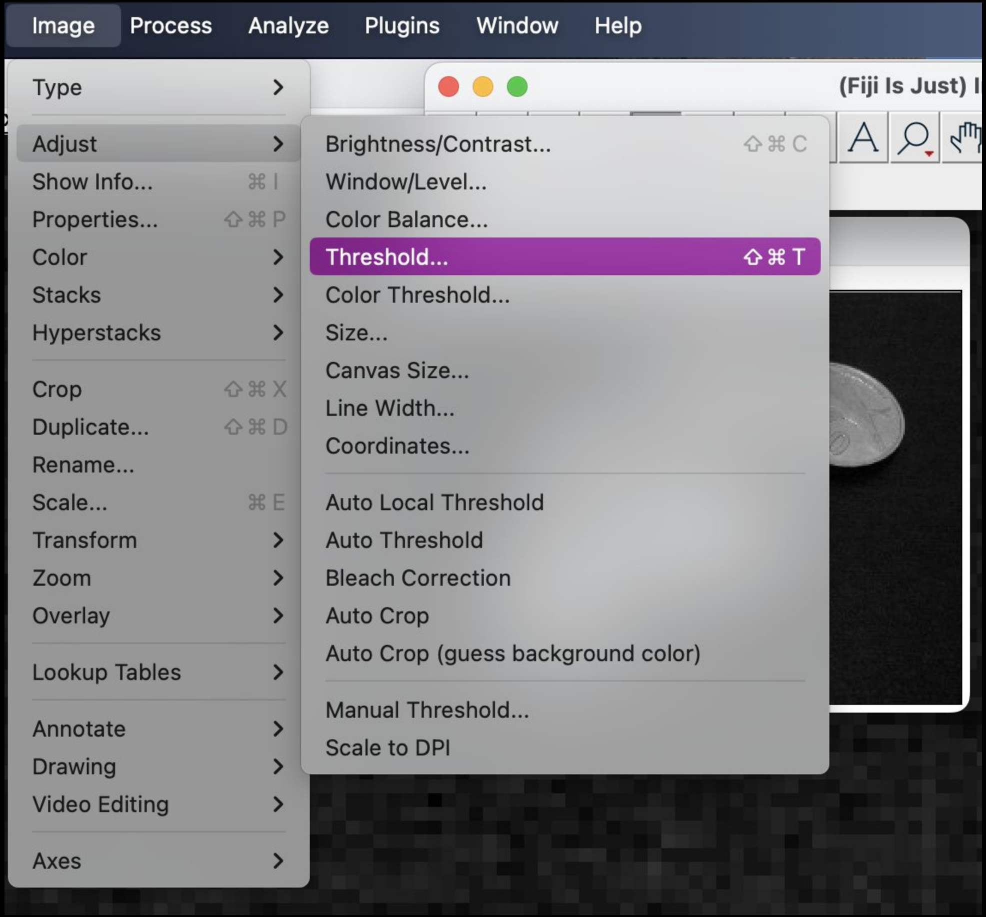


Threshold it

→Image → Adjust → Threshold

The default method finds the valley easily

Click Apply to make a **mask**



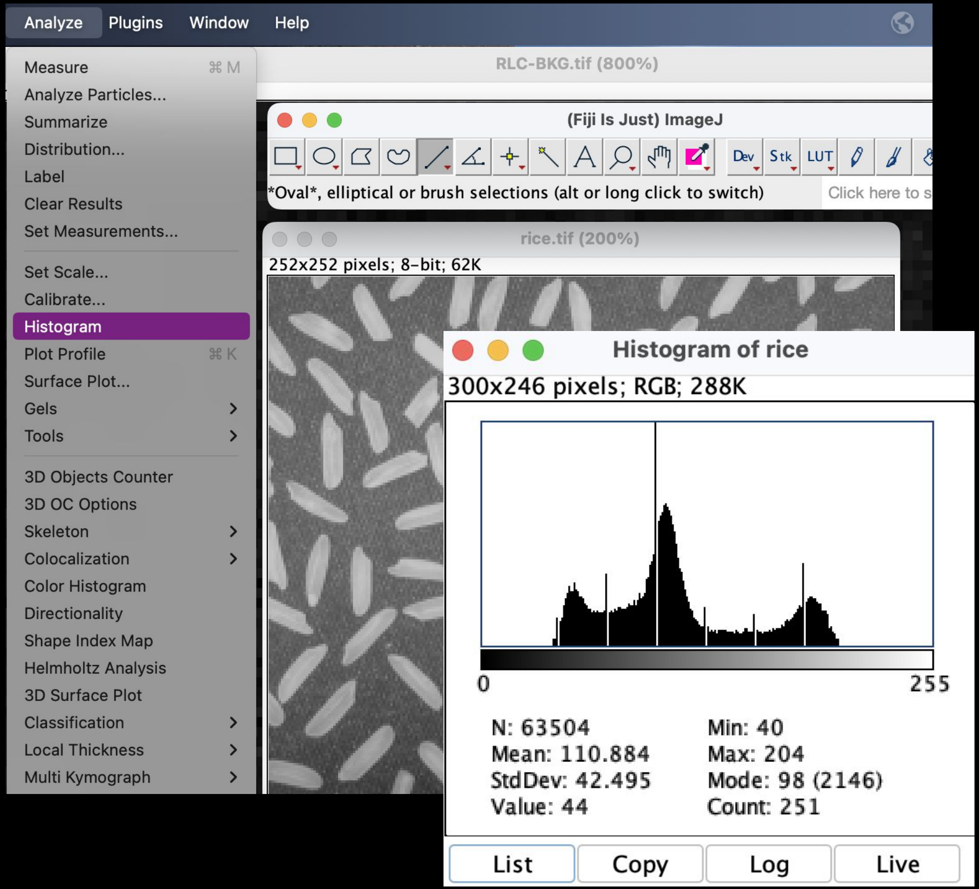
What is a Mask?

https://guide.cryosparc.com/processing-data/tutorials-and-case-studies/mask-selection-and-generation-in-ucsf-chimera

Try Rice Image

Open rice.tif from GitHub

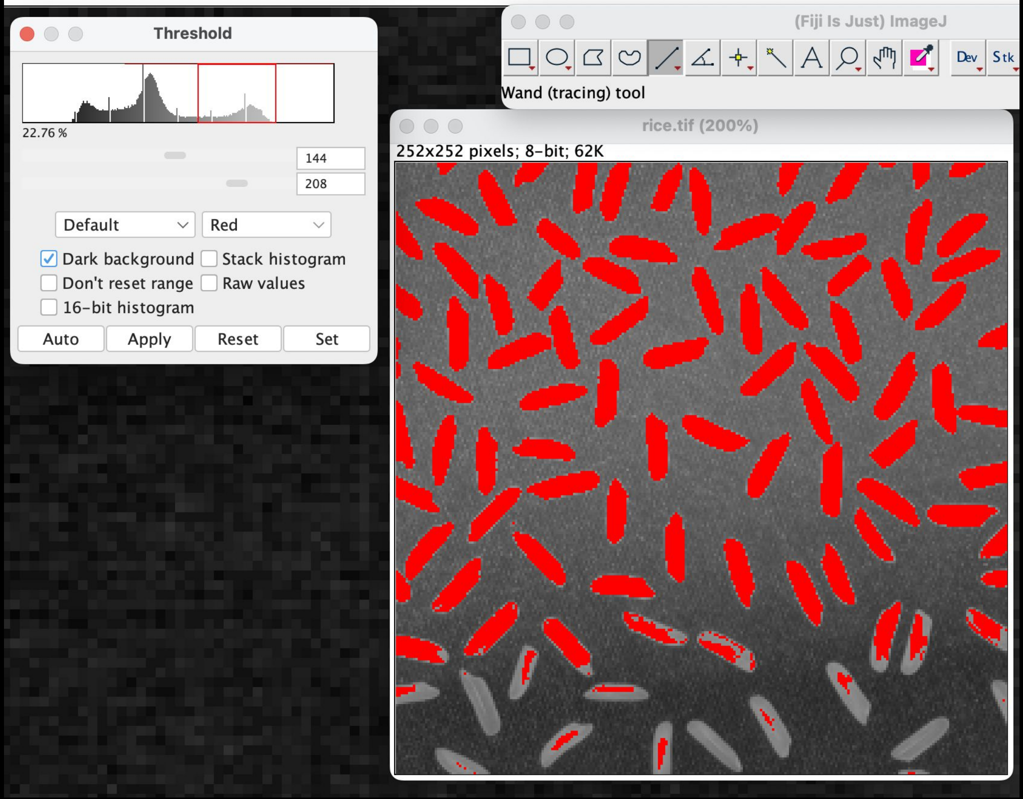
→ Analyze → Histogram



Creativity Is Essential

Try segmenting rice.tif

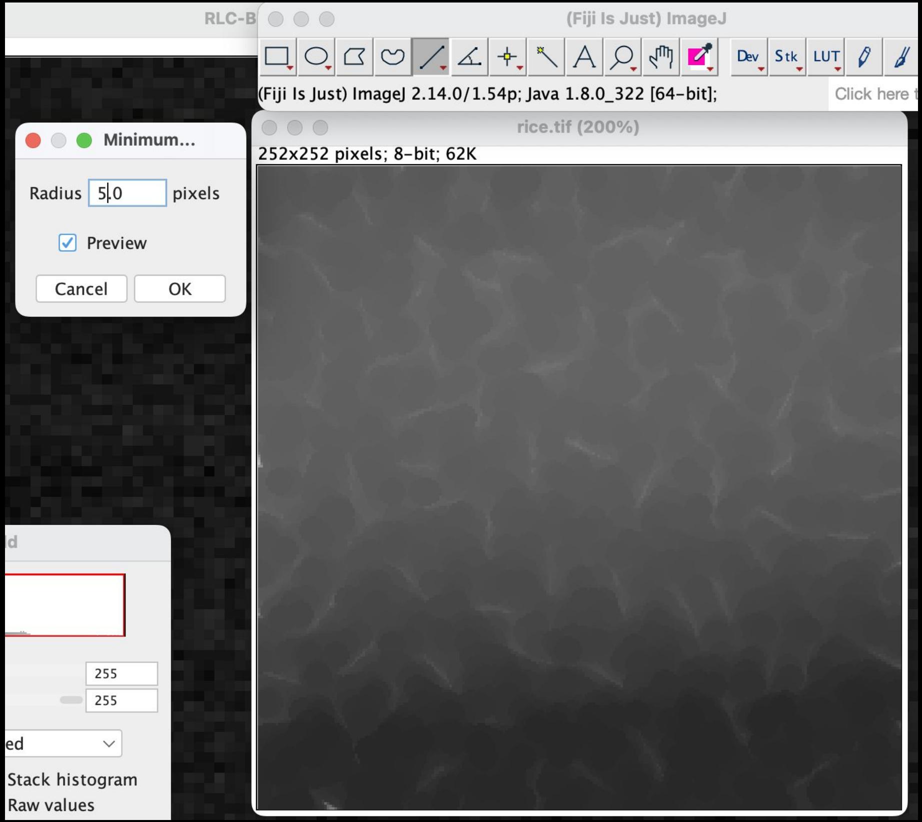
→Image →Adjust →Threshold



Try applying filters to improve segmentation

Image Math can save us

Step 1: minimum filter to remove rice completely (save image)



Step 2: Image Calculate

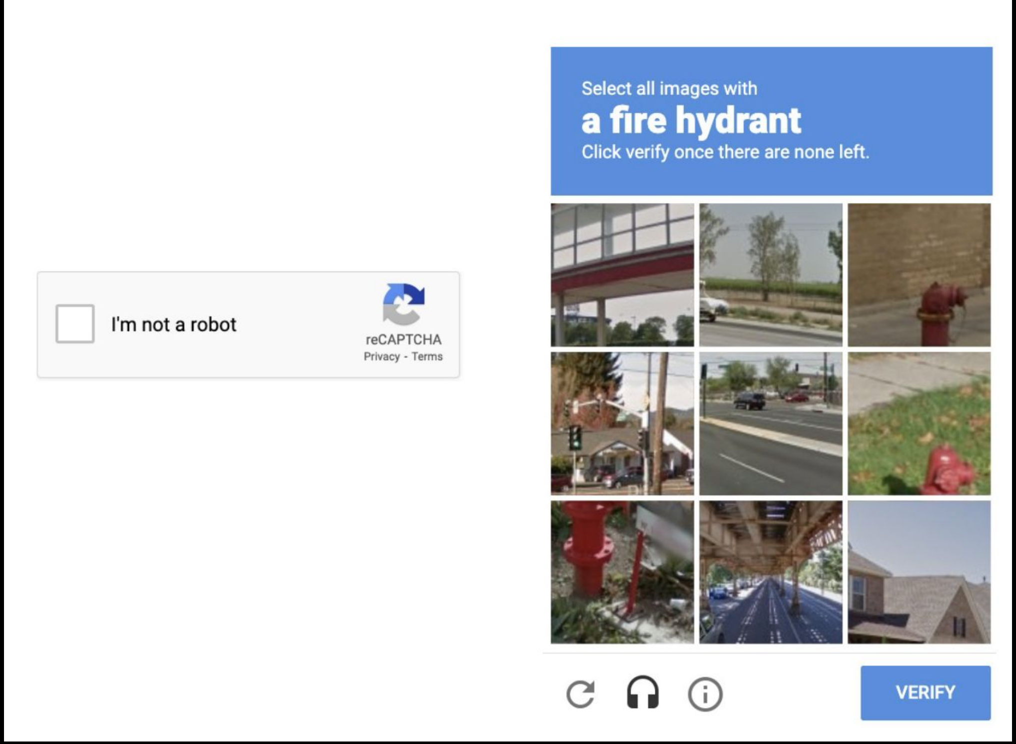
original - blurred image = segmentable image

What happened - a rolling circle replaced each pixel with the smallest pixel value in the neighborhood (radius 5) - erosion

Machine Learning for Segmentation

ML is teaching a computer to segment objects so that it can do it on its own

Essential for segmenting large image sets - do you want to go through this process by hand 10 times, 100 times, 1000 times?



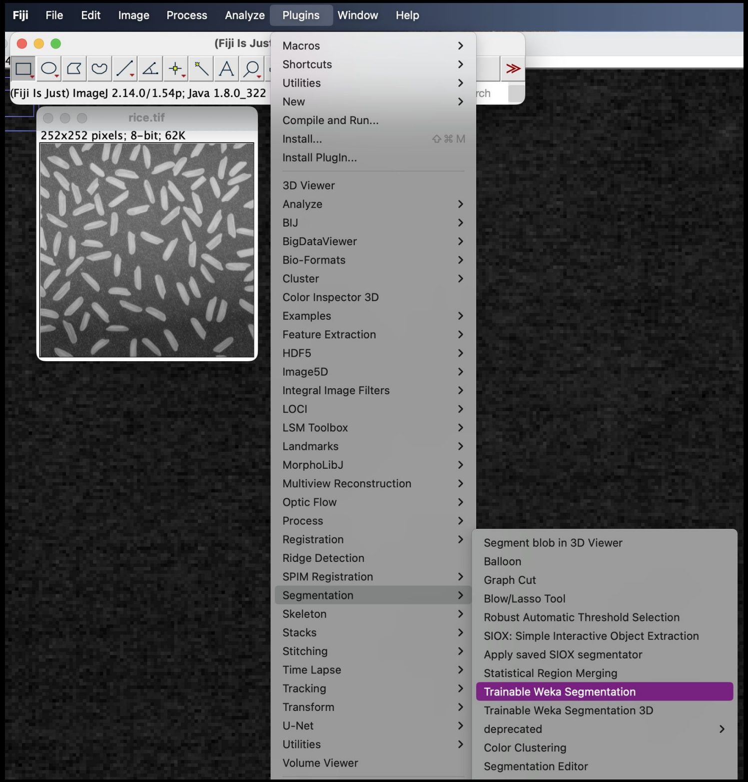
FIJI - Trainable Weka Segmentation

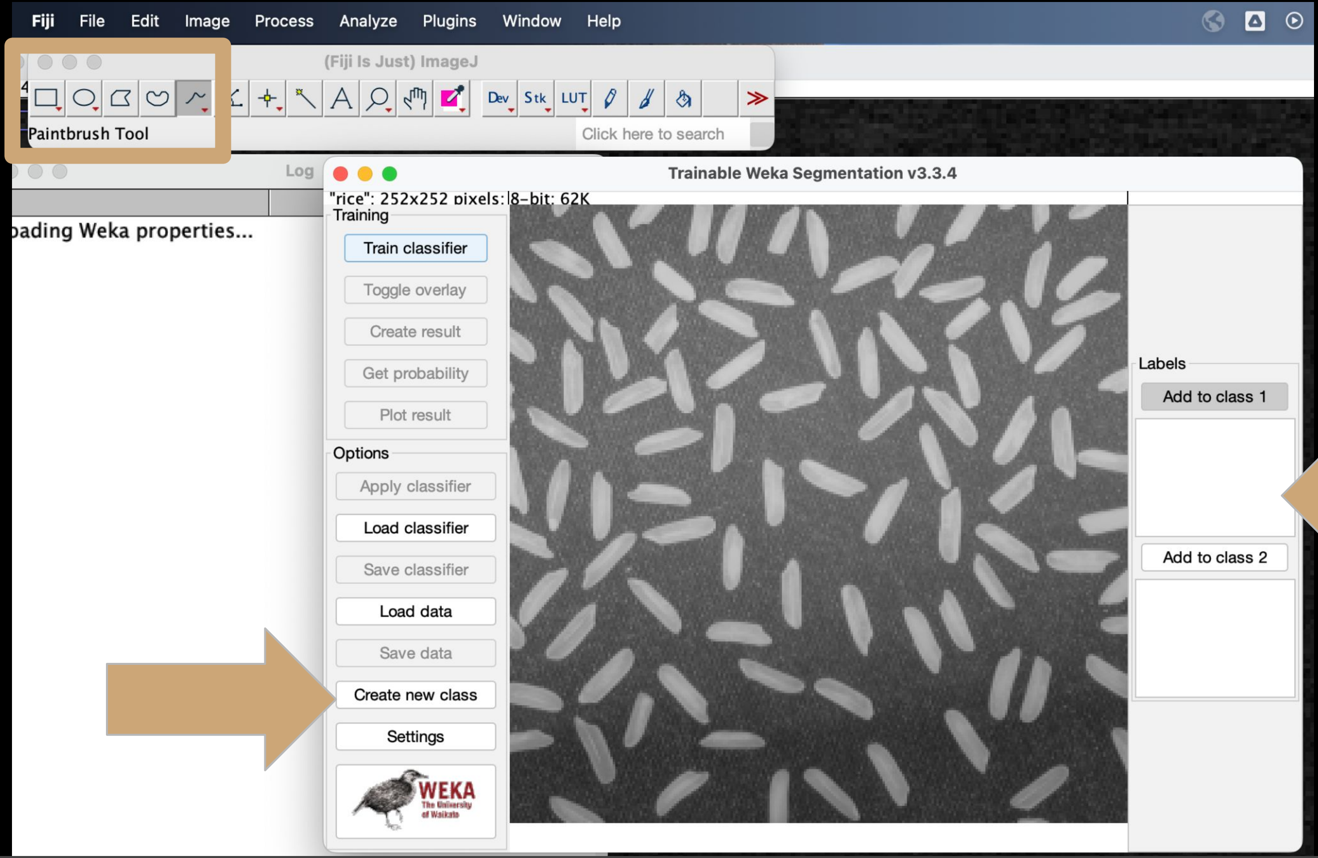
Open

→Plugins

→Segmentation

→Trainable Weka Segmentation

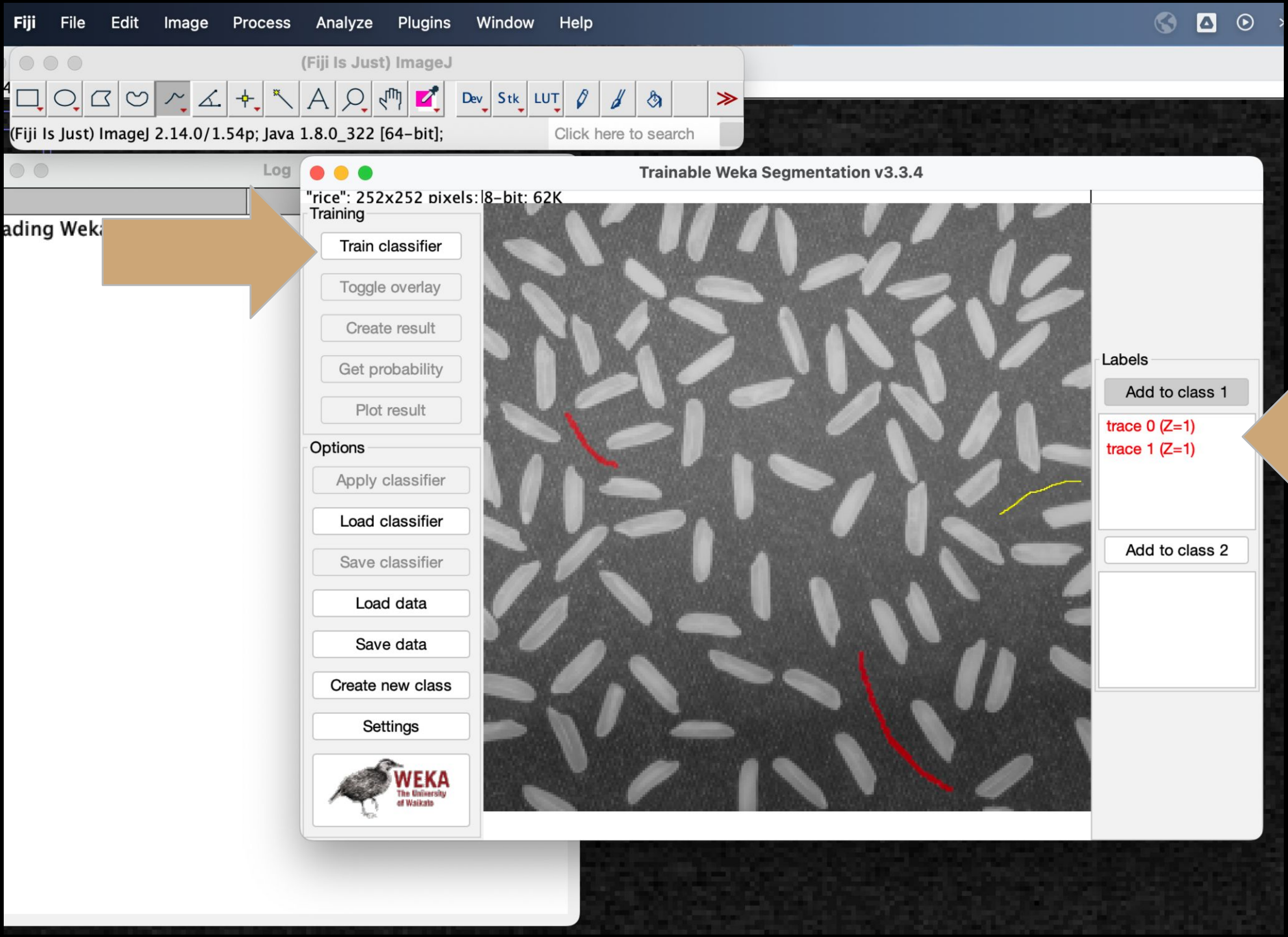




Classes are different categories for pixels

More classes can be added if needed

Classifiers are a type of ML algorithm that learns the classes and predicts - here we ask it to predict classes for pixels we have not identified



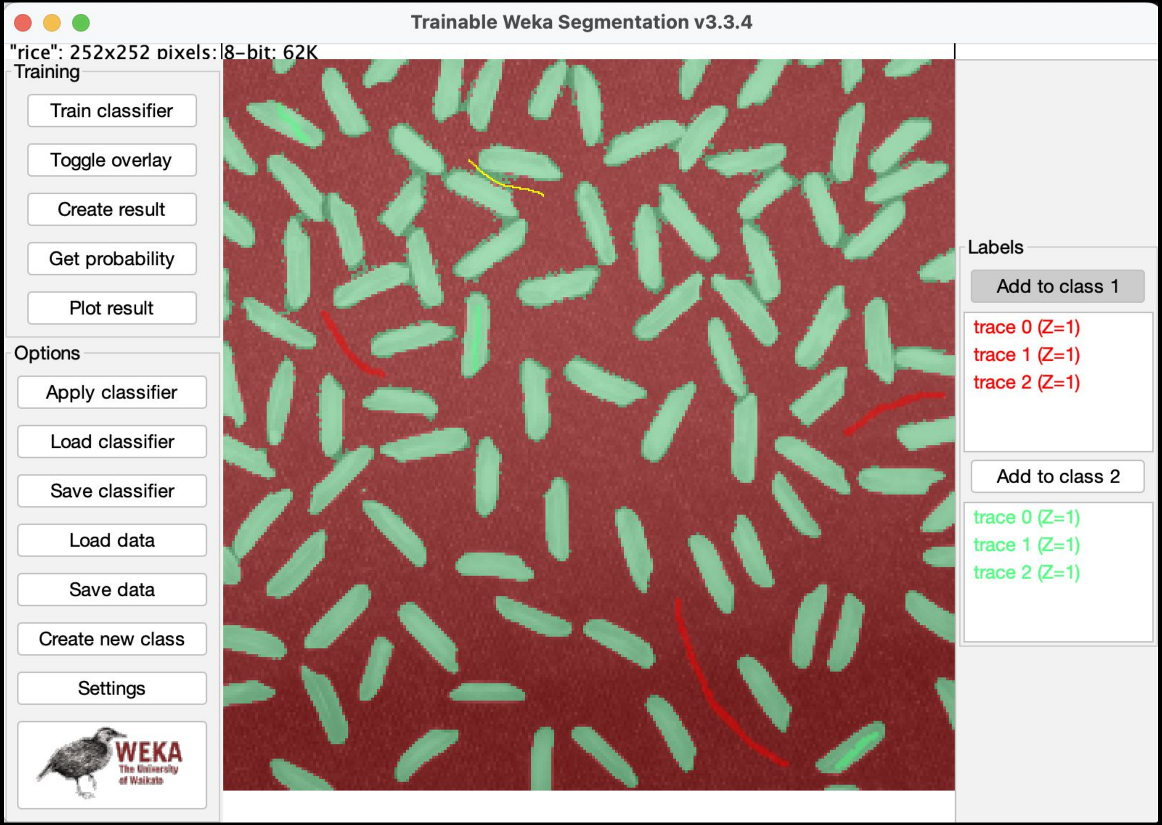
Draw a few shapes on the background

Add to class 1

Draw a few shapes on target objects

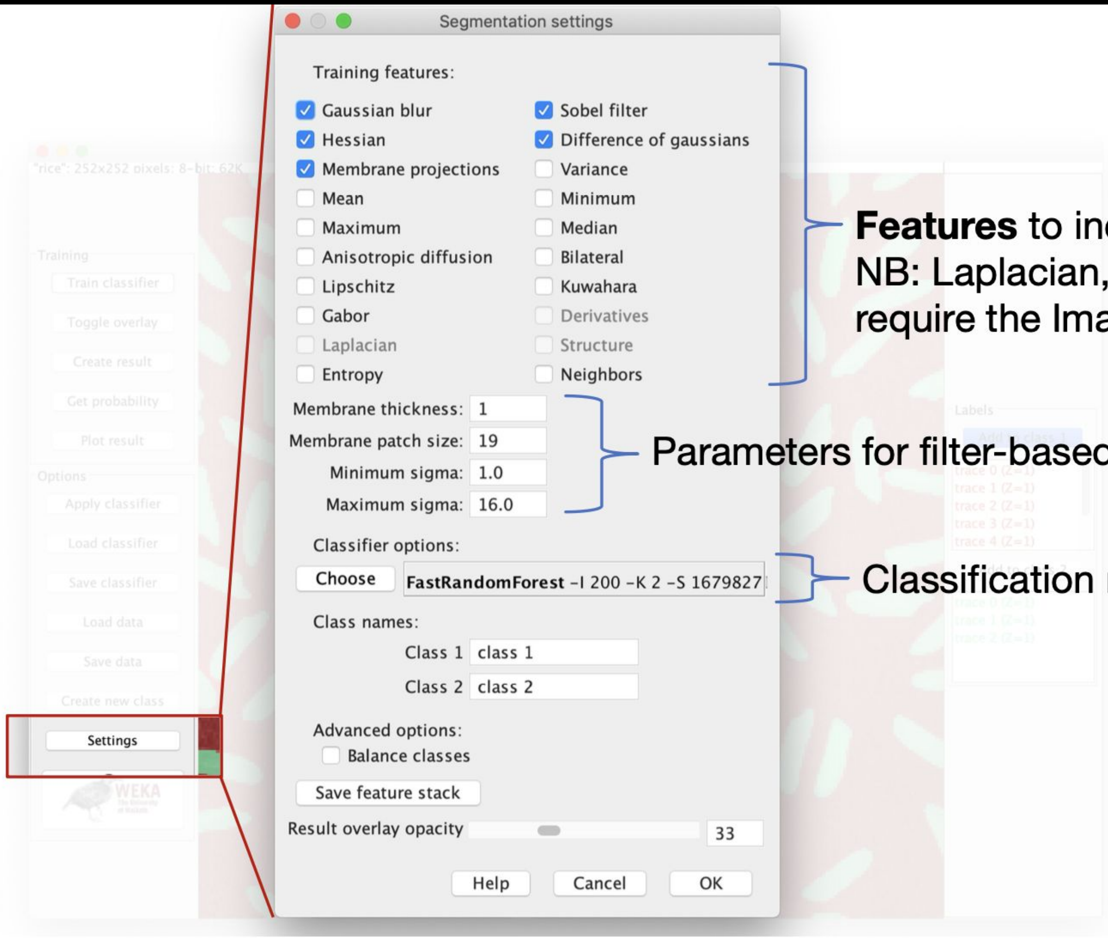
Add to class 2

Click 'Train classifier'



Add ROIs to improve segmentation

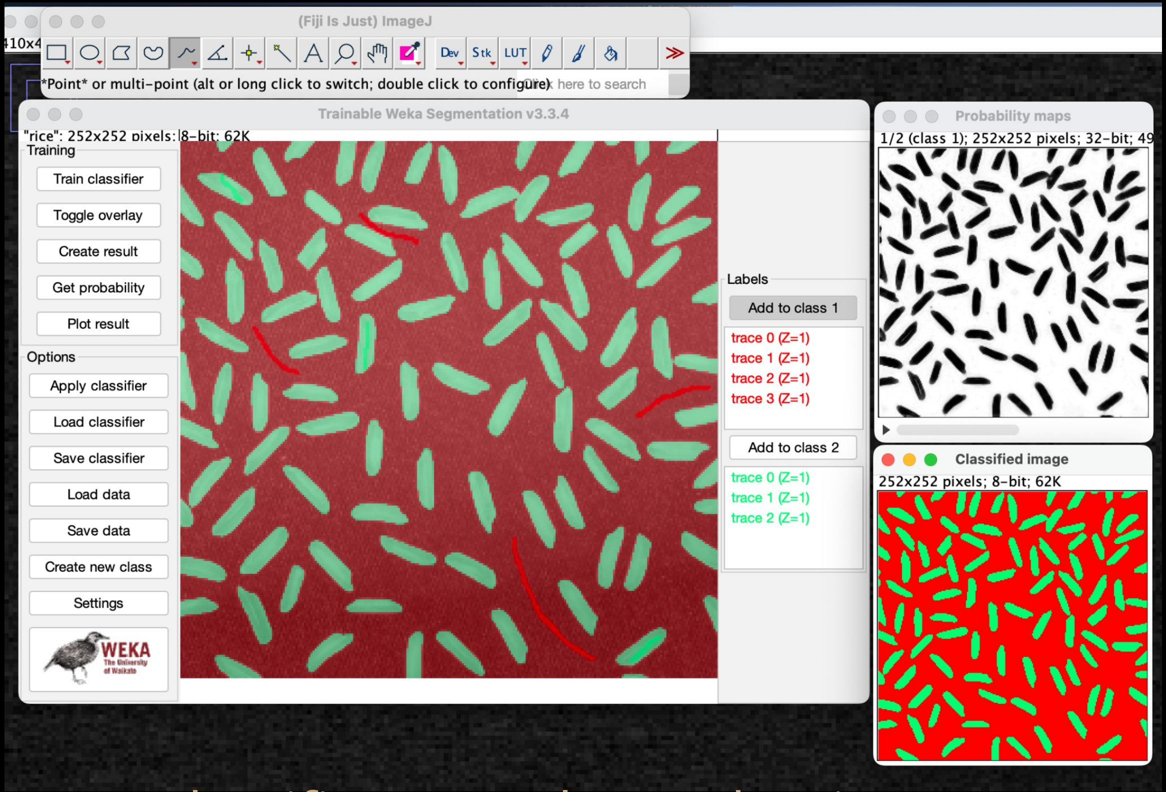
Less is better - don’t start with 20 ROI



Features to include in the **Feature Vector**
NB: Laplacian, Derivatives, and Structure require the ImageScience Update Site

Parameters for filter-based **Features**

Classification model



Get Probability - shows where it is unsure

Create Result - makes a label image

Now it can be thresholded for masked images

Save Classifier to apply to other images

Apply Classifier - will analyze an entire folder of images

Try another image

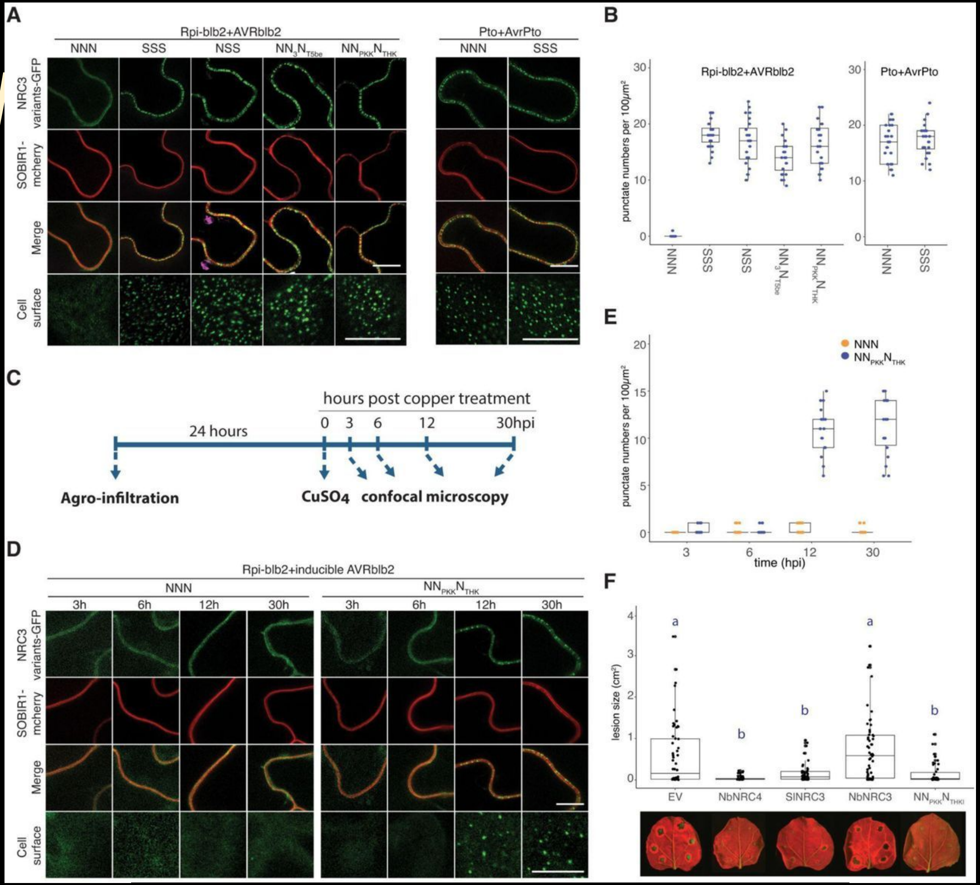
Your images, FIJI samples, or from our GitHub

So we segmented…now

The real goal: quantify stuff (to answer some scientific question)

Analysis

- Measure
- Count
- Track
- Co-localize



Huang et al., 2023

Let’s explore FIJI analysis tools

Take your time!

One puzzle at a time, be creative!

Microscopy Reporting

" I used 60X objective."

Numerical Aperture (NA)

Higher NA→higher resolution

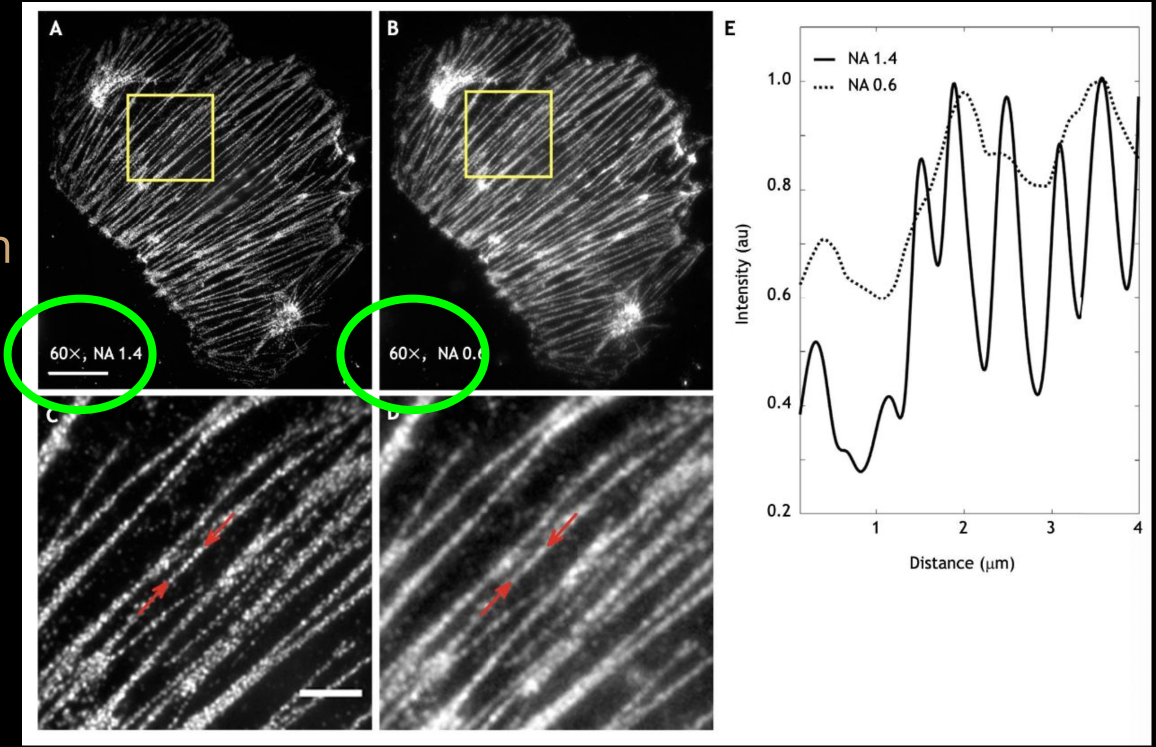
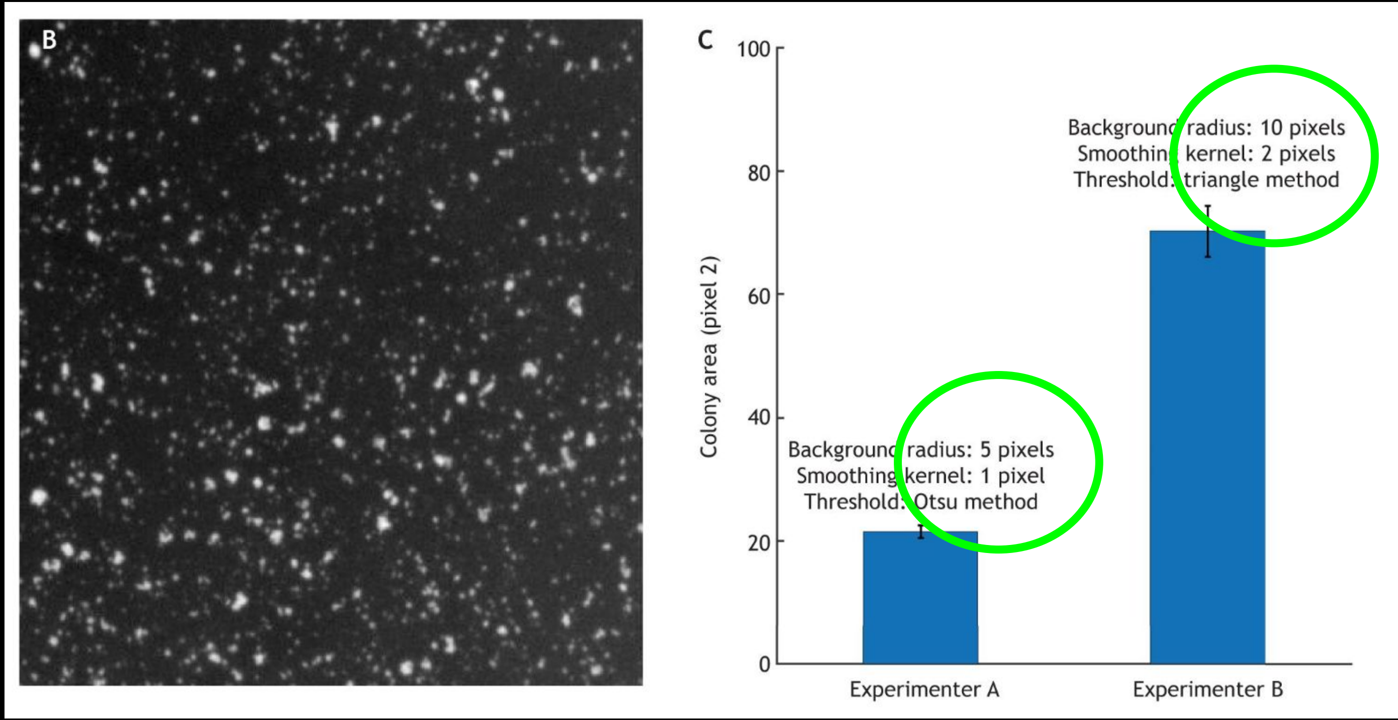


Image Processing Reporting

Setting details matter



PROCESSING TASK	COMMON ALGORITHMS	PARAMETER(S) TO REPORT
Background Subtraction	Rolling Ball/Rolling Minimum	Kernel size and shape
	Gaussian Smoothing and Subtraction	
	FFT high pass filter	Frequency cut-off values
Denoising	Gaussian Smoothing	Kernel size and shape
	Median Filtering	
	Non-Local Means	Noise sigma and smoothing value
Deconvolution	Weiner Deconvolution	PSF and Wiener factor (1/SNR)
	Richardson-Lucy	PSF and number of iterations
	Blind	Estimated PSF (or none)
Intensity Threshold	Manual	Intensity value or percentile
	Automatic	None
	Local	Varies
Segmentation	Pixel Connectivity	4 or 8 (2D images) 6, 18, or 26 (3D images)
	Morphometric Filtering	Shape/size parameter(s) with upper and lower bounds
	Binary Operations	Operations type (e.g. erosion, dilation), number and order of implementations

MICROSCOPE COMPONENT	PARAMETERS TO REPORT
Light source	Source type, make and model
	Wavelength (if laser)
	Power/Power Density at Sample
Excitation/Emission Optics	Dichroic mirror manufacturer and wavelength characteristics
	Excitation and Emission Filters manufacturer, and wavelength characteristics
	Instrument-Specific Components (e.g. confocal pinhole size)
Sample and Acquisition Parameters	Sample Preparation and Mounting Protocol
	Z-interval spacing
	Time-lapse Interval
Objective Lens	Manufacturer
	Magnification
	Numerical Aperture
	Optical Aberration Corrections
Detector	Detector Type and Manufacturer
	Exposure/Dwell Time
	Gain
	Offset
	Binning (if used)

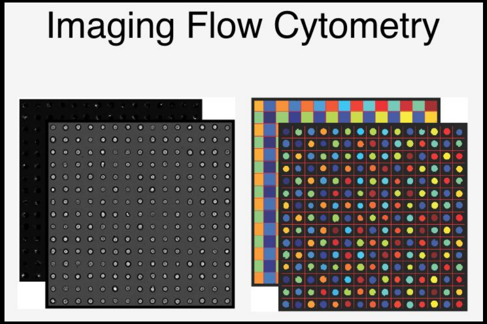
Reporting your processes

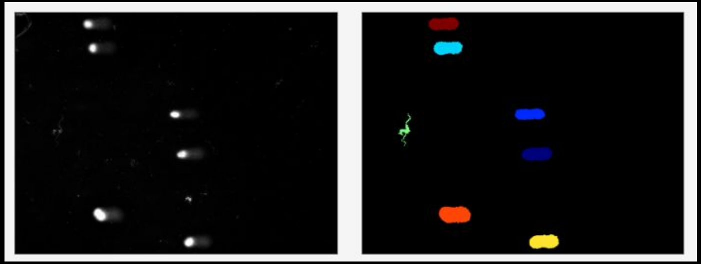
Aaron et al., 2021 A Guide to reporting in digital image processing - can anyone reproduce your quantitative analysis? https://www.aicjanelia.org/_files/ugd/835c50_cc34305735124c7092f2216e03b9bd5d.pdf

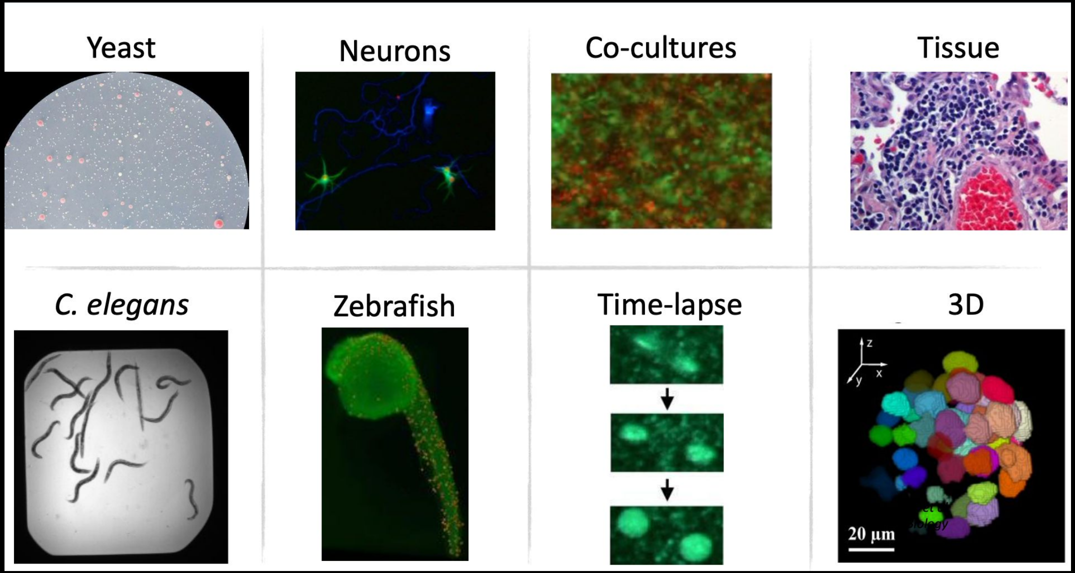
Heddleston et al., 2021, A guide to reporting in digital image acquisition - can anyone replicate your microscopy data? https://www.aicjanelia.org/_files/ugd/e43f7a_021d414cb2b14e24bbe65285d5938887.pdf

Other Software for Image Analysis

CellProfiler







- Friendly user interface
- Input nearly any kind of image
- Machine learning algorithms under the hood
- Automated - process thousands of images in minutes
- Exceptional support network for troubleshooting

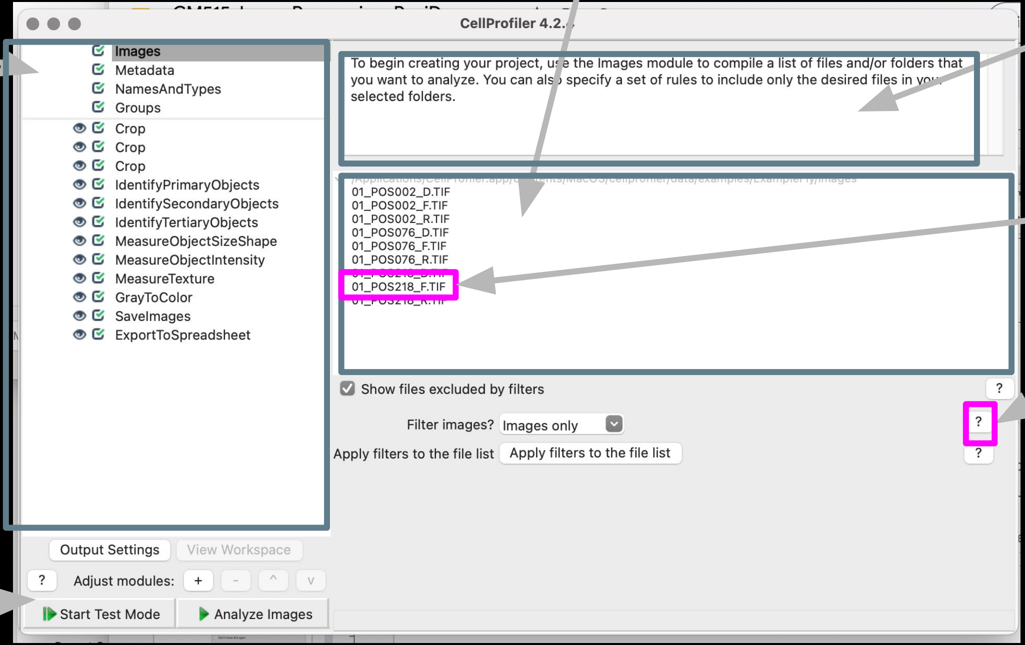
https://cellprofiler.org/

CellProfilerUser Interface

Images are loaded here, drag and drop or right click and select an option from the drop down

Information about each module: edit this so users know what to do

Module window: add or delete modules as needed to build your pipeline



Examine images double click

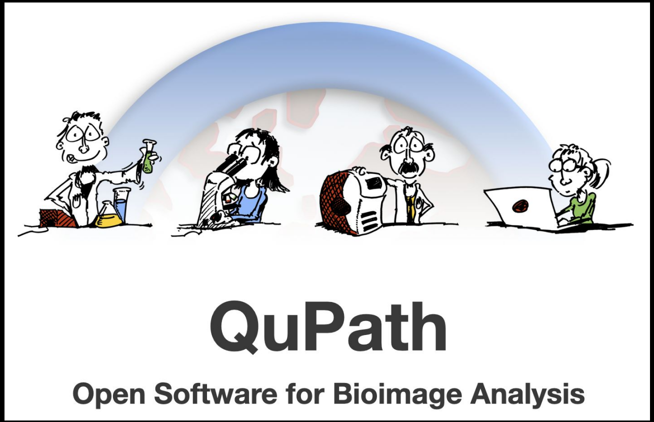
Use the ?

They are very helpful for learning what each function does

Tools for building and running the pipeline

QuPath Bioimage Analysis

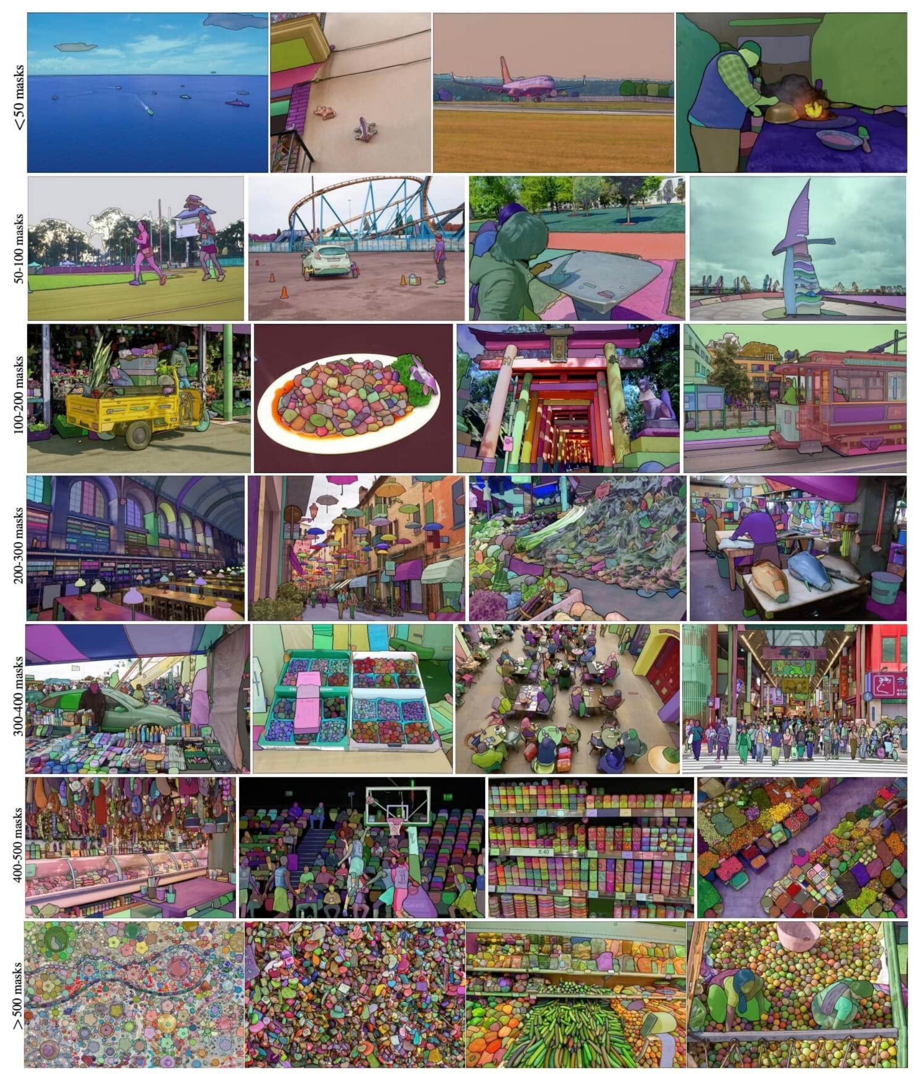
- https://qupath.github.io/
- Free Software
  - Mac OS
  - Windows



- Getting started
  - First steps
  - Viewing images
  - Annotating images
  - Manual counting
  - Two essential tips
  - Getting help
- Tutorials
  - Projects
  - Separating stains
  - Detecting tissue
  - Measuring areas
  - Cell detection
  - Cell classification
  - Density maps
  - Exporting measurements
  - Multiplexed analysis
  - Pixel classification
  - Superpixels

SAM-segment anything model

https://docs.ultralytics.com/models/sam/#introduction-to-sam-the-segment-anything-model



Free software for analyzing and processing image data

Fiji: ImageJ plus a large bundle of plugins, all packaged together in one easy-to-download application. Runs on Mac OS, Windows, and Linux. Very handy for new users who don't want to download individual plugins. Includes utilities for 3D viewing, video editing, measuring colocalization among others. **Recommended for all CDB Micro Core users.**

QuPath: Free, open-source software designed primarily for annotation and analysis of large-format whole-slide images (color brightfield and multi-channel fluorescence)

CellProfiler: Free, open-source software from the Broad Institute for segmentation and quantitative analysis of microscope image data.

Ilastik: Free software that uses machine learning for interactive object recognition and segmentation.

Imaris Viewer: A free version of Imaris with limited capabilities; useful for quickly viewing Imaris files or raw data.

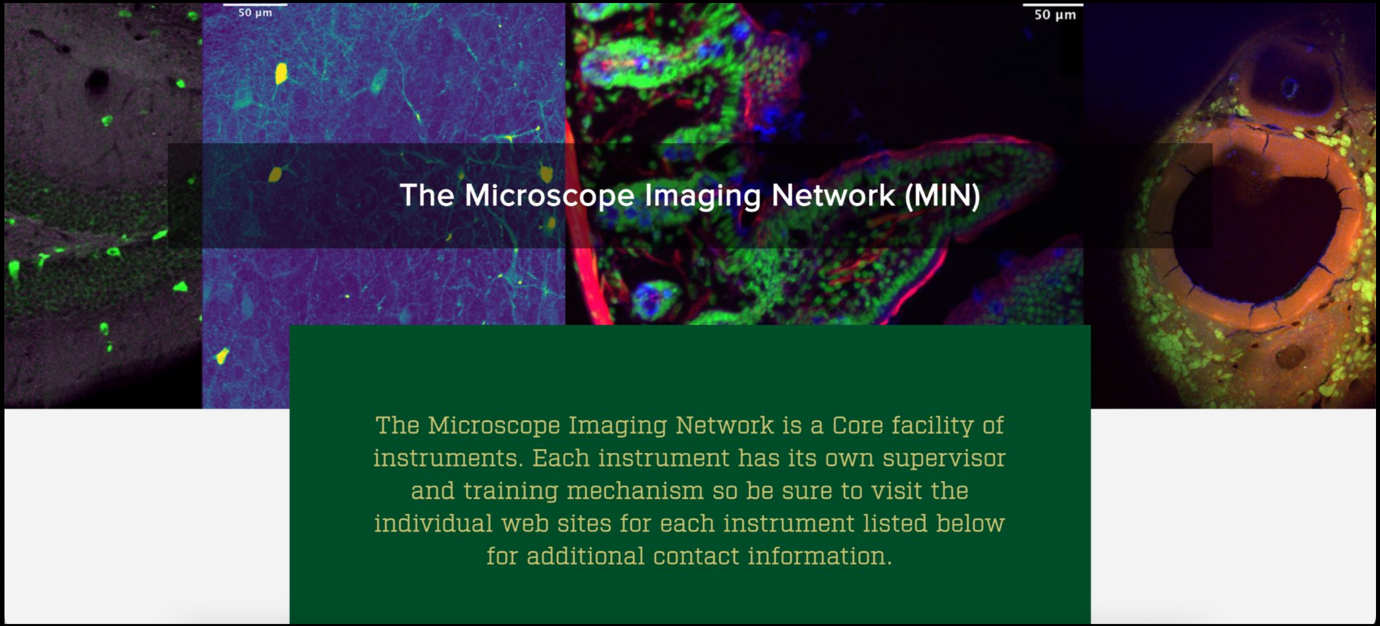
ZEN Lite : Free software from Zeiss to open .czi files. Windows only.

LAS X Core: Free software from Leica to open images acquired on the Leica SP8 confocal or any microscope controlled by LAS X. Scroll down the linked page to find the version appropriate to your operating system. Windows only.

On-Campus Resources

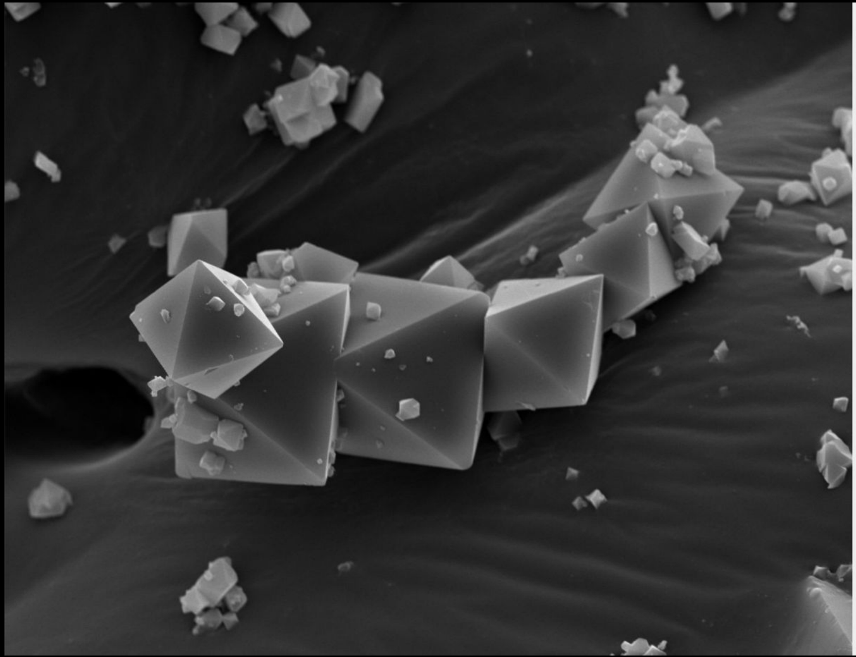
MIN Webpage

https://www.research.colostate.edu/min/



ARC-ISS Webpage

https://www.research.colostate.edu/iss/



Center for Imaging and Surface Science

The ISS Center enables research and development programs by providing expertise and access to state-of-the-art equipment for electron microscopy, spectroscopy, and other surface characterization measurements.

CuBTC metal organic frameworks. Dr. Jon Thai, Reynolds group, Chemistry Department, CSU

Courses Offered (qCMB electives)

NSCI 677	Microscopic Image Collection & Processing	2
BC 665A	Advanced Topics in Cell Regulation: Microscopic Methods	2
CS 510	Image Computation	4

Other Learning and Communities

MDI Biolabs spring Workshop - Quantitative Fluorescence Microscopy: https://mdibl.org/course/qfm-2025/

MBL fall workshop: https://www.mbl.edu/education/advanced-research-training-courses/course-offerings/optical-microscopy-imaging-biomedical-sciences

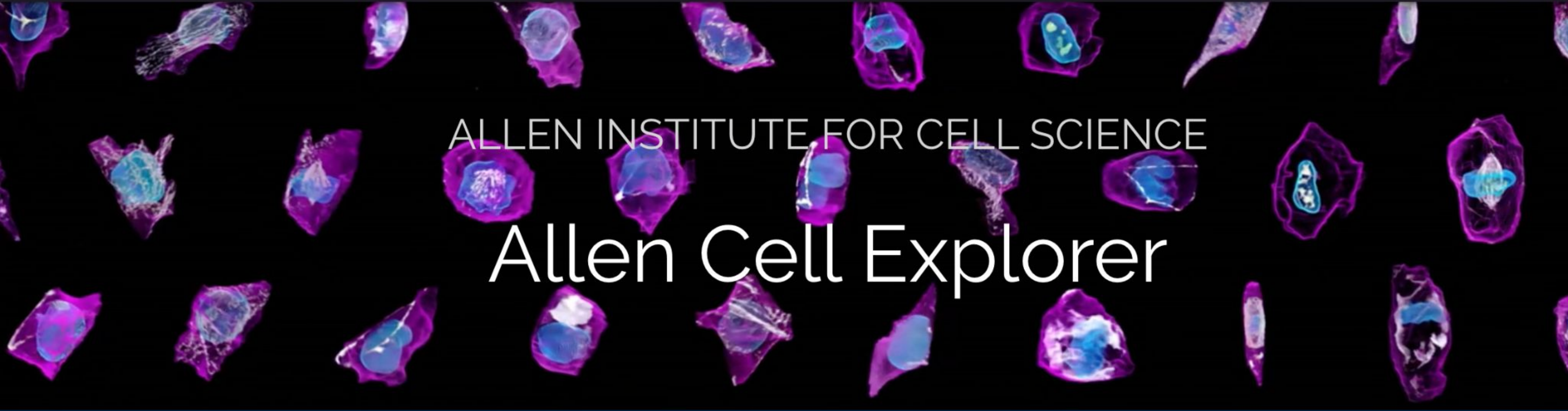
Cold Spring Harbor Lab spring workshop: https://meetings.cshl.edu/courses

BINA community: https://www.bioimagingnorthamerica.org/join/

Public Image Databases

ALLEN INSTITUTE for CELL SCIENCE

About | Allen Cell Collection | Data & Digital Tools | Analysis & Modeling | Publications | Education | Support



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Images

Biological Images

- Acute Lymphoblastic Leukemia Image Database
- Allen Brain Atlas
- American Society of Hematology Image Bank
- Blood Cell Images Dataset
- Broad Bioimage Benchmark Collection
- Cell Image Library
- Euro-BioImaging FAIR-Sharing Collection
- Malaria Cell Images Dataset
- Microscope Cell Nuclei Images Dataset
- Molecular Expressions Photo Gallery
- Multimodal Videos of Human Spermatozoa

Challenge Images

- BigNeuron Gold166 Benchmarking Images
- DIADEM Neuron Reconstruction Challenge
- Fingerprint Verification Competition
- Grand Challenges in Biomedical Image Analysis
- ImageNet Large Scale Visual Recognition Competition
- PASCAL Visual Object Classes Challenge